

MAXIMISING EXAM MARKS IN **Senior maths**

INTERMEDIATE Book 2

*Top tips for
avoiding
common errors!*



BARBARA LYNCH



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Introduction

Every year, examiners, teachers and assessors in every state produce detailed reports on how students have coped with the various questions on their exams. And every year the same errors are made by students. *Maximising Exam Marks in Senior Maths* is a collection of reproducible worksheets which are designed to assist senior students in those areas that they have traditionally found difficult.



Questions focusing on the common errors in the context of the topic area in which they typically occur will help students avoid wasting valuable marks. Key labels mark important content, concepts and techniques:

THINGS TO KNOW – Content that students should learn and be able to apply

HINTS – Helping students approach questions correctly

TRICKS – Giving short-cuts or easier ways to attack questions

SLIPS – Providing opportunities for students to practise working carefully to avoid common errors

TRAPS – Providing the types of contexts in which students make common errors

The table of contents below shows the topic areas used in each activity. Pages which stress appropriate examination and revision techniques are also included. The worksheets could be integrated into the learning process or used as part of examination or test revision. In most cases, an electronic calculator would be useful and where appropriate, a graphical calculator can be used.

The CD-ROM attached to the inside back cover contains a PDF version of this book.

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1 – Key words and getting them right

Watch out for some key words and they often show you how to do it!

Make sure you read the question carefully before answering it and know your rules!



- Equation will probably mean solve to get a number
- Factorise means express as product of factors
- Simplify means make simpler
- Express usually gives you the form of the answer
- Single fraction means common denominator
- Hence means use what you have already done
- Evaluate means find the value of
- Exact value means keep as surds or other exact numbers
- Show means list and explain all the steps.

1 A Express as a single fraction in simplest form:

$$\frac{(3x-2)}{3} - \frac{(x+2)}{4} = \frac{4(3x-2) - 3(x+2)}{12}$$

$$= \frac{\quad}{12}$$

$$= \frac{\quad}{12}$$

B Hence solve exactly $\frac{(3x-2)}{3} - \frac{(x+2)}{4} = 1$.

From Part (a): $\frac{\quad}{12} = 1$

$\times 12$ both sides $\quad = 12$

Simplify to get $x = \quad$

2 A Simplify $\frac{a^2-1}{a}$.

B Express $\frac{a^2-1}{a} = 2$ as a quadratic equation and hence find the exact value of a .

C Find the value of a correct to 2 decimal places such that $\frac{a^2-1}{a} = 2$.

D Factorise $\frac{a^2-1}{a}$.

3 A Simplify (make simpler) $\frac{2}{x+5} - \frac{1}{x}$.

TRAP

Be careful of the signs when removing brackets.

TRAP

This is an equation to solve, and 'hence' means use what you have already done. Note that the answer must be exact.

HINT

Divide denominator into numerator.

HINT

Use your calculator to convert surds.

HINT

Factorise the numerator.

TRICK

Keep the brackets on for as long as possible! Note these words: simplify, solve, hence, evaluate, exactly.



1 – Key words and getting them right (cont'd)

Ⓑ Hence evaluate (find the value of) $\frac{2}{x+5} - \frac{1}{x}$ when $x = 5$.

Ⓒ Solve $\frac{2}{x+5} - \frac{1}{x} = 2$ for x **exactly** using the quadratic formula.

HINT

No 'hence' but question follows one part on another! Also, it tells you how to do it!

THINGS TO KNOW

Know the rules!

Ⓓ Show by describing each step how the equation $2\sqrt{x+7} = 3x$ can be simplified to $(9x+14)(x-2) = 0$ and state what restrictions need to be placed on the value of x and why.

Step 1 _____ :

Step 2 _____ :

Step 3 _____ :

Step 4 _____ :

Ⓔ Algebraic manipulation causes problems for many students, and it is helpful to practise identifying what method can best be used to factorise algebraic expressions.

- **CF** is common factor.
- **DOPS** is difference of perfect squares.
- **DOCS/SOCS** are difference and sum of cubes.
- **FTh** (Factor Theorem).

THINGS TO KNOW

Know the rules!
Always common factor first!

Copy and complete the following table.

	Expression	Number of terms	Method	Factorisation
Ⓐ	$x^3 - 27y^3$	2	DOCS	$(x-3y)(x^2+3xy+9y^2)$
Ⓑ	$a^2 - 9b^2$	2	DOPS	
Ⓒ	$x^4 - 9x^2$	2	CF & DOPS	$x^2($
Ⓓ	$2x^2 + 3x - 2$	3	Quadratic	
Ⓔ	$x^3 - 6x^2 + 8x$	3	CF & Quadratic	$x($
Ⓕ	$3x^3 - 5x^2 + x + 1$	4	FTh	$(3x+1)($
Ⓖ	$x^4 - 2x^3 - 5x^2 + 6x$	4		$x(x-1)(x-3)(x+2)$
Ⓗ	$x^3 - 4x^2 + 2x - 8$	4	Group in pairs	
Ⓘ	$(a-b)(a+b) - (a-b)^2$	2	CF	
Ⓙ	$3x^2 + x - 10$	3		
Ⓚ	$x^3 - 4x^2 + x + 6$	4		
Ⓛ	$(x+2)^2 - (2x+1)^2$	2		

Ⓜ Spelling – There is nothing that upsets an examiner more than poor spelling of mathematical terms and incorrect punctuation and grammar. Here are some words to learn to spell – copy them out: factorise, coordinate, statistic, equation, Cartesian, quadratic, probability, discriminant, differentiate, algebraic, antidifferentiate, trigonometric.

2 – Tricks and traps in integration

The idea that integration is the reverse of differentiation leads to some standard results.

Different examinations provide different formula sheets, but you should still learn them off by heart.

The best way to do this is to practise.



- ① Many of these standard integrals will be on your list to learn. Cross out the ones you don't need and write down in your own notation the ones you need to learn.

THINGS TO KNOW

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

where $n \neq -1$ and $x \neq 0$ if $n < 0$

$$\int (ax + b)^n dx = \frac{1}{a(n+1)} (ax + b)^{n+1} + c$$

where $n \neq -1$ and $x \neq 0$ if $n < 0$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + c, \text{ for } a \neq 0$$

$$\int \frac{1}{x} dx = \log_e x + c, \text{ for } x > 0$$

Anti-differentiation by parts:

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + c, \text{ for } a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + c, \text{ for } a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + c, \text{ for } a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + c, \text{ for } a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + c, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + c, a > 0, -a < x < a$$

$$\int \frac{-1}{\sqrt{a^2 - x^2}} dx = \cos^{-1} \frac{x}{a} + c, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + c, x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + c$$

Note: $\ln x = \log_e x, x > 0$

$$\int \frac{f'(x)}{f(x)} dx = \ln f(x)$$

- ② The antiderivative of $\frac{2x}{x^2 + 1}$ is $\log_e(x^2 + 1) + C$. State *an* antiderivative of $\frac{2x}{x^2 + 1}$.

HINT

The words 'antiderivative', 'primitive' and 'integral' are often used interchangeably. 'An' integral means any one of them but 'the' integral means a definite one, so you need to add '+ c'. Remember that $\sqrt{x} = x^{\frac{1}{2}}$.

- ③ Find the antiderivative.

A $3x^2 - 3x^3 + 18x - 5$ _____

B e^{15x} _____

C $\sin 15x$ _____

D $e^{3x} - \frac{3}{x+1} + \sqrt{x}$ _____

- ④ Find a primitive of $2 - \sec^2 x$. _____

- ⑤ Find the following integrals.

A $\int \frac{15}{x} dx$ _____

B $\int \frac{1}{x^2} dx$ _____

- ⑥ Find:

A $\int \sqrt{3x} dx$ _____

B $\int \frac{2}{x-3} dx$ _____

C $\int (-3x^2 - 8x) dx$ _____

- ⑦ Find $\int (3x-1)(3x^2-2x)^2 dx$ but do **not** simplify the answer.

TRAP

Careful! Algebra integrals sometimes give log answers and sometimes algebra answers!

TRAP

Be careful to antidifferentiate not differentiate, watch for the log ones and remember the + c. Remember that $\sqrt{3x} \neq \sqrt{3} \times x$.

TRICK

Multiply out first!

3 – Definite integrals

Antidifferentiating and then substituting are the skills you need to find the value of a definite integral. Often you need to find exact rather than decimal values, so a calculator can't be used. Of course you can always check your answer with a calculator. Remember that not all functions are zero when you substitute the value zero.



- ① Find the exact value.

A $\int_0^2 2x^5 \cdot dx$ _____

B $\int_0^1 e^{5x} \cdot dx$ _____

- ② Complete the following definite integral by simplifying after the limits are applied.

$$\int_0^{\frac{\pi}{3}} (\cos 2x) \cdot dx = \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{3}} = \left(\frac{1}{2} \sin 2 \frac{\pi}{3} \right) - \left(\frac{1}{2} \sin 0 \right) =$$

- ③ Find the exact value of a for which $\int_0^a 3e^{3x} \cdot dx = 4$. Write your answer in the form $\log_e k$ where k is an integer.

④ If $\int_0^1 \frac{1}{2x+1} \cdot dx = \left[\frac{1}{2} \log_e(2x+1) \right]_0^1 = \log_e a$, find a .

- ⑤ Find $\int_1^2 (e^{2x^2}) \cdot dx$ correct to three decimal places by entering $\int (e^{(2 \cdot x^2)}, x, 1, 2)$ into your graphical calculator.

There is no need to show your working. _____

- ⑥ A Show that $\frac{1}{x-4} + \frac{-1}{x-1} = \frac{3}{x^2-5x+4}$.

B Use this fact to find p where $\int_5^6 \frac{3}{x^2-5x+4} \cdot dx =$
 $\int_5^6 \left(\frac{1}{x-4} - \frac{1}{x-1} \right) \cdot dx = \left[\log_e(x-4) - \log_e(x-1) \right]_5^6 = \log_e p$

TRAP

Substitute zero carefully.
Exact value not decimal value.

HINT

Know exact values of
trigonometric functions.

HINT

Treat a as if it were just a number
at first and make sure you show
your working and know your log laws!

HINT

Remember the log laws when you
substitute the integral's limits.
For what values does the log exist?

TRAP

Watch the brackets!

TRAP

Watch for algebraic and arithmetic
errors. Perhaps draw a graph to see
what the maximum looks like.



3 – Definite integrals (cont'd)

HINT

Remember the log laws when you substitute the integral's limits.

- 7 A Find $\int_1^3 (x-1)^2 dx$ by first multiplying out the bracket.

HINT

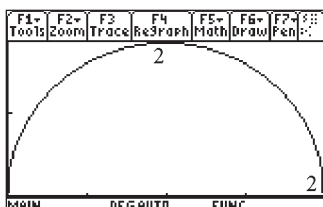
When does an integral give an area?

- B Hence find the area under the curve with equation $y = (x-1)^2$ and between the line with equation $x = 1$ and the line with equation $x = 3$. Explain your answer.

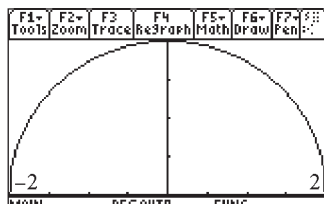
HINT

Show the meaning of everything in your representation.

- 8 On the graph of the semicircle $y = f(x)$ for $0 \leq x \leq 2$ shown, draw a representation of $\int_0^1 f(x) dx$.



- 9 The graph of $y = \sqrt{4-x^2}$ for $-2 \leq x \leq 2$ is shown. Find the exact value of $\int_{-2}^2 \sqrt{4-x^2} dx$ by the easiest method.



TRICK

Integral problems are not always area problems and area problems aren't always integral problems!

HINT

Semicircle

- 10 If $R(t) = 6 + e^{0.4t-2}$ is the rate of usage of paint in thousands of litres per year, find the amount used in the first 10 years. Give your answer to the nearest thousand litres.

TRAP

Note that $R(t)$ is a rate already, so the amount used is found using the integral! Remember to substitute $t = 0$ as well as $t = 10$.

- 11 A Find the value of k such that $\int_{-6}^k (-x^2 - 6x) dx = 0$.

HINT

Watch for algebraic and arithmetic errors. Perhaps draw a graph to see what the maximum looks like.

- B Find the value of k such that the value of $\int_{-6}^k (-x^2 - 6x) dx$ is a maximum.

4 – Integration and area under the curve

Sometimes the integration gives the actual area under a curve and sometimes it gives the signed area. The easiest way to make sure you get it right is to sketch a graph (on the graphical calculator if you are allowed to use it) and look to see if the graph goes below the x -axis. If it does then you need to divide up the area into sections. Remember, sometimes integration is just a calculus skill.



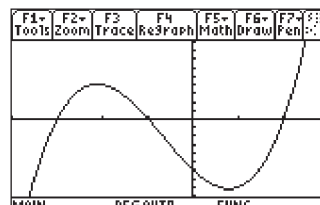
- ① Which of the following questions has an answer that has units?

What are those units? _____

Ⓐ Evaluate $\int_0^2 \frac{1}{x^2 + 1} dx$.

- Ⓑ Find the area under the curve with equation $y = \frac{1}{x^2 + 1}$ between $x = 0$ and $x = 2$.

- ② For the graph with the rule $y = (x + 3)(x + 1)(x - 2)$ shown, find the area between the curve and the x -axis.



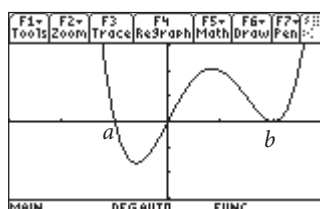
TRAP

Did you get a numerical answer? If so, re-read the question and answer the question asked.

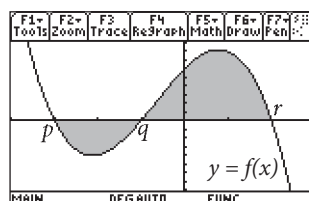
HINT

Decide whether to use the absolute value (modulus) of the integral or subtract areas. Multiply out carefully before trying to integrate. Also watch for arithmetic errors – keep all the brackets on.

- ③ For the graph of $y = f(x)$ shown, will $\int_a^b f(x) dx$ be greater than, less than or equal to the area between the graph and the x -axis between $x = a$ and $x = b$? Give reasons for your answer.



- ④ State the formula you would use to find the area between the graph of $y = f(x)$ and the x -axis in the interval $[p, r]$; that is for $p \leq x \leq r$.

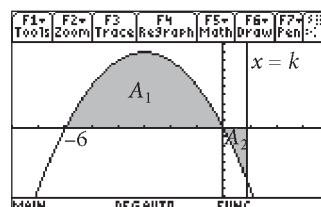


TRAP

Make sure you write in the dx in the integral sign. You want the actual area so be careful! Decide how to handle the area below the x -axis.

- ⑤ The graphs with rules $y = -x^2 - 6x$ and $x = k$ are shown.

- Ⓐ Find A_1 .



- Ⓑ If $\int_{-6}^k (-x^2 - 6x) dx = 0$, state A_2 and explain your answer.

HINT

To explain you should discuss the values of the integrals not the areas.

- ⑥ The graph with equation $y = x^3 - 3x + 2$ has one x -intercept at $x = -2$.

- Ⓐ Find any other intercepts using a graphical calculator or by using the Factor Theorem:

$$x^3 - 3x + 2 = (x + 2)(x^2 \quad \quad) = (x + 2)(x \quad \quad)(x \quad \quad)$$

x -intercepts are at $x = -2$, $x = \quad \quad$ and $x = \quad \quad$



4 – Integration and area under the curve (cont'd)

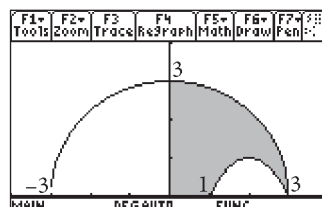
- ⓑ Explain your answer.

- ⓒ Hence find the exact value of the area enclosed by the graph and the x -axis.

HINT

Explain is best done in words.
Draw a graph to help this and
the next part of the question.

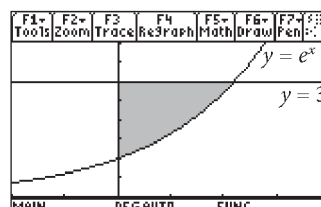
- ⑦ The graphs of $y = \sqrt{9 - x^2}$ and $y = -x^2 + 4x - 3$ are shown.
If the area of the quarter of a circle is $\frac{9\pi}{4}$, write down an expression
involving an integral that you could use to find the shaded area.



TRAP

The exact value should be a logarithm.

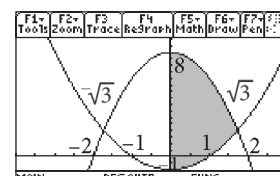
- ⓑ ⓐ For the graphs of $y = e^x$ and $y = 3$ shown, find the exact value
of x for which they intersect.



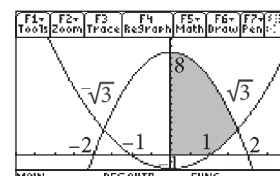
- ⓑ Hence write down the integral that would be used to find
the shaded area.

- ⑨ The graphs of $y = x^2 - 1$ and $y = 8 - 2x^2$ are shown.

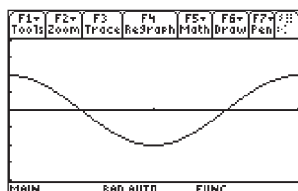
- ⓐ Write an integral expression that could be used to find the shaded area in this graph.



- ⓑ Write an integral expression that could be used to find the shaded area in this graph.



- ⑩ The graph of the function $f(x) = \cos 2x$ is shown for $0 \leq x \leq \pi$.
Find the area, correct to 3 decimal places between the curve
 $y = f(x) - 1$ and the x -axis.



HINT

- ☞ Draw the translated graph and shade the area you want.
- ☞ Decide what terminals you will need on the integral.
- ☞ Decide what expression you will need to integrate.
- ☞ Show all your working and be careful how you use your calculator!

5 – Rates of change in calculus

The rate of change of one variable with respect to another can be thought of as a measure of the gradient of the graph of their relationship. Rate questions are easy to identify because the word 'rate' is usually in them somewhere! In worded problems you must use the correct units.



THINGS TO KNOW

When no 'with respect to' is mentioned, the rate of change of a variable is a measure of the gradient of the graph of the relationship with time. So for $y = f(x)$, the rate of change of y with respect to x is $\frac{dy}{dx}$ and the rate of change of y is $\frac{dy}{dt}$.

The average rate of change between two points is the measure of gradient of a secant or chord of the graph between those points.

- ① Find the average rate of change of the curve with equation $y = (2x - 1)^2 + 1$ between the points where $x = -1$ and $x = 2$.

- ② For $y = \cos t$, where t has the units of time, find the exact value of the average rate of change of y over the interval $\left[0, \frac{\pi}{2}\right]$; that is, $0 \leq t \leq \frac{\pi}{2}$.

- ③ Find the average rate of change of y with respect to x for the curve with equation $y = 2^x$ between $x = 2$ and $x = 2.1$, correct to 2 decimal places.

- ④ If N is the number of spores present after time t hours, find the average rate of change of $N = 50e^{0.2t}$ in the first 5 hours. What is the rate of change of N at 5 hours? Give your answers correct to two decimal places.

- ⑤ Find the rate of change of y with respect to x for $y = \log_e x$ at:

A $x = 4$.

B $x = b$.

TRAP

- Note that average rate means gradient of chord.
- There is no mention of 'with respect to' so it must be with respect to time.
- Be careful that you have your calculator in radians not degrees, but you are required to give an exact answer.

HINT

Gradient of the chord or gradient of the tangent.



THINGS TO KNOW

Terms to know:

- ☞ 'Rate of increase of a function' means the value of the function is getting larger and you want the gradient.
- ☞ Likewise, 'rate of decrease of a function' means the value of the function is getting less but you still need to find the gradient (which will be negative).

Remember: Gradient function is $f'(x)$.

- 6 Find the rate of increase of the function with rule $f(x) = \frac{12x + 2x^2 - 15}{x}$.

TRICK

Divide by x first!

- 7 What is the instantaneous rate of change of volume after 5 seconds if $V(t) = 3\cos 3t + t$ where t is in seconds and V is in m^3 ? Give your answer correct to 3 decimal places, with units.

TRICK

Make sure you are in radian mode on the calculator since time is the variable!

- 8 If $y = |\sin(x)|$, what is the rate of change of y with respect to x at $x = k$, where $\pi < k < 2\pi$?

HINT

What does the graph look like?

- 9 Which of the following questions has an answer that has units, and what are those units?

A Find the rate of change of position x metres after time t seconds where $x = 3t + 5t^2$.

B Find the rate of change of x where $x = 3t + 5t^2$.

HINT

Did you work out an answer? Maybe it doesn't ask for a number!

- 10 The radius of a circular oil slick is increasing at the rate of 0.2 m/s. After calculation, it is found that at 100 m radius, the rate at which the area of the oil slick is increasing is given by $\frac{dA}{dt} = 40\pi$.

A What are the units of this rate of change of area?

B Give the rate of change of area at 100 m radius correct to 2 decimal places.

C Give the rate of change of area at 100 m radius correct to 2 significant figures.

6 – Rates of change in practice

Mathematical formulae describe the relationship between variables. If a formula is being used for a practical situation, then the variables in the formulae will have units. If the formula is not part of a practical problem, then no units are necessary. Remember that units are used for the comparison of measurements of quantities but they are also useful in identifying what is given or required. Metric units of some form are normally used in Mathematics problems. Of course you must read all questions carefully.



- ① If 1 L = 0.22 gallons and 1 km = 0.62 miles, my car's petrol consumption is 15 miles per gallon. Use a calculator to convert this to L per 100 km.

TRAP

Oops! One consumption is miles over gallons, but we want litres over kilometres. Need to invert fraction.

- ② The equation $V(t) = 0.01t^2 - 2t + 100$ gives the volume $V \text{ cm}^3$, of unmelted ice-cream in a tub, t seconds after it has been taken out of the freezer. What is the rate (in cm^3/sec) at which the ice-cream is melting, 20 seconds after it has been taken out of the freezer? Explain your answer.

HINT

Meaning of negative rate.

- ③ The rate of usage of a particular chemical in an experiment in time t hours is given by $R(t) = \frac{5e^{0.2t}}{5 + e^{t-10}}$ grams per hour.

TRAP

Be careful of algebraic errors and use the calculator for checking the calculus only.

Remember, R is the rate already and you want the maximum rate, so you want the maximum value of R .

- A Use calculus to find the value of t for which the rate of usage is a maximum for $0 \leq t \leq 20$.

- B What is the maximum rate of usage?

- ④ A 5000 L water drum is to be emptied. The volume of water in the tank, V litres, after time t minutes is given by $V = 5000\left(1 - \frac{t}{90}\right)^2$; $0 \leq t \leq 90$.

TRAP

Watch the units in the questions and answers! Note the rate graph will be a straight line, so check the ends of the domain. Think about a likely answer.

- A Use substitution to find the volume of water in the tank after 1 hour, correct to 2 decimal places.

- B After how long will the tank be empty?

- C Find the rate at which the water is flowing out after 1 hour, correct to 2 decimal places.

- D When is the fastest rate of water flow?



6 – Rates of change in practice (cont'd)

- 5 The equation $P = 125 + a \sin 10t$ gives the pressure of blood in a patient's arteries in the units mm of Mercury after t seconds of a units of activity.

(A) What is a suitable domain for t ? Why?

(B) Find the rate of change of pressure at time t .

(C) If the rate of change of pressure must not exceed 600 mm/s, find the maximum value of a .

- 6 The depth of water in a canal, h metres, at time t hours after midnight is given by $h = 5 \cos \frac{\pi t}{6} + 4$.

(A) Find an expression for the rate of change of the water level at time t .

(B) Hence find the rate of change of water level at 2 a.m. correct to 3 decimal places. Is the water rising or falling?

(C) Use a graph to estimate t (to the nearest hour) when the rate of change is first greatest.

- 7 The needle on a sewing machine moves up and down according to the rule $y = 30 + 25 \sin \left(\frac{(4t-1)\pi}{2} \right)$; $t \geq 0$ where y mm is the distance of the needle from the base at time t seconds from the start.

(A) Show that the rate of change of y with respect to t in simplest form is given by $50\pi \sin(2\pi t)$. What are its units of measurement?

(B) Find the value of rate of change of y with respect to t when $t = \frac{1}{4}$ correct to 1 decimal place.

- 8 A small object was oscillating up and down at the end of a vertical spring. The object was h metres above its starting point at time t seconds, where $h = e^{-3t} \cdot \cos 2t$ where $t > 0$.

(A) Find an expression for the rate of change of the height.

(B) Find the rate at which the height above the starting point is changing 1 second after motion started. Answer correct to three decimal places.

HINT

- Look at a graph for the correct domain for t .
- Note that t is in seconds, so don't have π in the answer.
- 'Must not exceed' means 'less than or equal to'.
- Be careful of the differentiation. Which value of t will give the greatest value of a ?

HINT

Practise differentiating trigonometric functions, check units and put the calculator in radian mode. Notice t is time after midnight.

TRAP

A common error is to omit pi or give an answer with pi in it.

HINT

Even if you couldn't do the 'show' in part (a) you should still attempt part (b) using the given expression.

HINT

Notice you need the product rule so apply it carefully. When substituting you need to be in radian mode. The answer must be correct to 3 decimal places. What does a negative answer mean?

7 – More than a rate

Some rates are related. When one variable changes, other variables that are related will also change. Often the chain rule will be useful if rates are related. So, for example, if $y = f(x)$ and $x = g(t)$ then $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$ will give the rate of change of y with respect to t in terms of the other two rates. These questions require a special approach using function of a function. Be careful to read the question to make sure that what looks like a rate is actually a rate type of question. Maximum and minimum questions can be hidden inside or added to a rate question.



- ① The depth of water is x cm in a jug containing v cm³ of water. Water is poured into the jug at a constant amount per second. What units could be used to describe:

TRAP

Units!

(A) the rate at which the volume of water in the jug is changing? _____

(B) the rate at which the level of water is changing? _____

(C) the rate of change of volume of water with depth? _____

- ② The radius, r cm, of a sphere is increasing at a constant rate p cm per second. The value of the rate of increase of the volume of the sphere, V cm³ per second, is $4\pi r^2$ times the rate of increase of its radius.

(A) Complete the following.

(i) $\frac{dr}{dt} =$ _____

(ii) $\frac{dV}{dt} =$ _____

HINT

Letters work like numbers but you can't use a calculator.

- (B) Hence find the exact value of the rate of increase of the volume of the sphere at the instant when the radius is 3 cm.

THINGS TO KNOW

READ

- ☞ What rate is required; say, $\frac{dy}{dt}$?
- ☞ What rate is given; say, $\frac{dx}{dt}$? Is it increasing (positive) or decreasing (negative)?
- ☞ Write a chain rule to connect the rate required and the rate given; here, $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$.

PREPARE

- ☞ What is the linking derivative; here, $\frac{dy}{dx}$?
- ☞ Now you need to find a relationship between them so you can differentiate to find the link in the chain rule; here, $y = f(x)$

DO

- ☞ Differentiate $y = f(x)$ to give $\frac{dy}{dx}$.
- ☞ Substitute all the links you have into your chain; here, $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$.
- ☞ Solve to give the rate you need.

CHECK

- ☞ Answer the question asked: does the question ask for the value of the rate or something else.
- ☞ Check that your answer is feasible and sensible.

REREAD TO CHECK YOU'VE FINISHED THE QUESTION

- ☞ Tick off each part of the question to be sure you have answered it.

- ③ $y = 200x^{\frac{3}{2}}$ and $\frac{dy}{dt} = 10$.

(A) Find $\frac{dy}{dx}$. _____

(B) Write a chain rule that could be used to find $\frac{dx}{dt}$. _____

THINGS TO KNOW

The chain rule doesn't necessarily have everything the right way up! $\frac{dy}{dx} = 1 \div \frac{dx}{dy}$



C Hence find an expression for $\frac{dx}{dt}$. _____

D Find the exact value of the rate of change of x when $t = 9$.

- 4 The volume of water, $v \text{ cm}^3$, in a jug is given by $v = 108x + x^3$ when the depth of water in it is $x \text{ cm}$. Water is poured into the jug at a constant rate of $30 \text{ cm}^3/\text{s}$. At what rate is the level of water rising when the depth is 8 cm ?

READ

A What rate is required? _____

B What rate is given? Is it positive or negative (increasing/decreasing)?

HINT

Units are a hint to help work out what derivative you need.

C Write the chain rule connecting Parts (a) and (b). _____

PREPARE

D What derivative do you need to connect the known rate with the unknown rate? _____

E What relationship will you use to find this? _____

DO

F Differentiate to find the connecting link in the chain rule. _____

G Substitute in the chain rule and solve for the rate required.

CHECK

H State the units of the required rate. _____

- 5 A melting snowball which is always spherical in shape is decreasing in volume at a constant rate of $12 \text{ cm}^3/\text{min}$. Find the rate at which the radius is changing when the snowball has a radius of 6 cm . Give your answer correct to 3 decimal places. (The volume of a sphere, V , of radius r is given by the formula $V = \frac{4}{3}\pi r^3$.)
- _____
- _____

TRAP

Careful! Positive or negative?

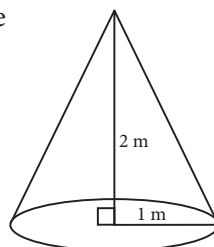
THINGS TO KNOW

'Right' means upright so the vertex is perpendicularly above the centre of the base.

- 6 A container in the form of a right cone with base radius 1 m and height 2 m is being filled with oil at a constant rate of $0.3 \text{ m}^3/\text{min}$.

A The volume of a cone, V , of radius r and height h is given by the formula $V = \frac{1}{3}\pi r^2 h$. Show that $V = \frac{\pi h^3}{12}$.

B Hence find at what rate the water is rising when the depth is 0.5 m , correct to 2 decimal places.



HINT

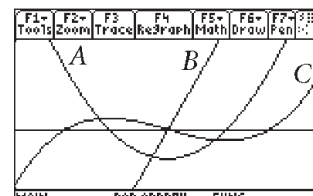
What do you substitute? Watch out for the 'show' and 'hence'.

8 – Moving particles

It is important to understand that kinematic relationships are not just about differentiating and antidifferentiating. The calculator can help with 'what' but you need to understand 'why'. Remember that if you are using a formula in a practical situation then bits of the formula will have units. But if a formula is not part of a practical problem, then no units will be necessary.



- ① For the formula $T = 2\pi\sqrt{\frac{l}{g}}$, which of the following questions has an answer that has units, and what are the likely units?
- Ⓐ Find the length l of a pendulum with period 0.5 s if the acceleration due to gravity g is 9.8 ms^{-2} .
- Ⓑ Find the value of l if $T = 0.5$ and $g = 9.8$.



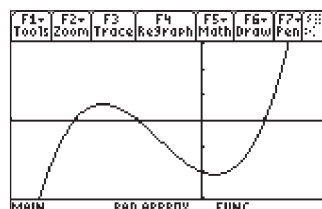
- ② The three graphs show the relationships between displacement and time; velocity and time; and acceleration and time. Which is which?

- Ⓐ _____
- Ⓑ _____
- Ⓒ _____

HINT

Looks hard, but you will find it is actually pretty easy if you read the question carefully!

- ③ The acceleration of a mass on the end of a spring in cm per sec² at time t sec after the start is given by $a = -5.4\pi \sin\left(\frac{3\pi}{2}t + \frac{\pi}{4}\right)$. Write the equation that could be solved to find how long after the start the mass will first have an acceleration of 0.5 cm per sec^2 .



- ④ The graph of the function with equation $x^3 + 2x^2 - x - 2$ is shown. Sketch the gradient function of this curve and provide a brief explanation of the relationship between the two graphs at the points where the gradient function crosses the x -axis.

THINGS TO KNOW

Explaining means making something clearer and **giving reasons**.

- ⑤ Explain the meaning of $\int_{20}^{40} v(t)dt$ in terms of the motion of the particle for the graph drawn from the table of values of velocity v metres per second at time t seconds.

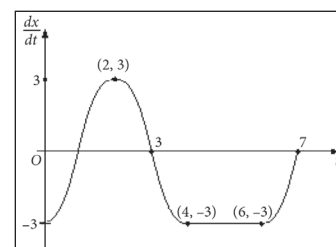
HINT

'Explain'

Time t seconds	0	10	20	30	40
Velocity v metres per second	0	20	15	40	60

- ⑥ The graph shows the velocity, $\frac{dx}{dt}$, of a particle as a function of time. Initially the particle is at the origin, O .

- Ⓐ At what time is the displacement, x , from the origin a maximum?



- Ⓑ At what time(s) does the particle return to the origin? Justify your answer.

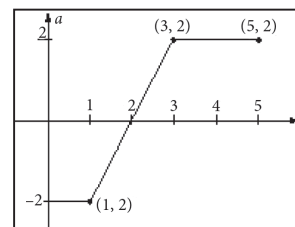
HINT

Make sure you put on a scale on the t -axis and be careful that the horizontal parts of the solution are clearly different from the t -axis. The curved (concave) sections should relate to the original graph.



- Ⓒ On a separate sheet of paper or on the same axes, draw a sketch of the acceleration, $\frac{d^2x}{dt^2}$, as a function of time for $0 \leq t \leq 7$.

- 7 A particle that is initially at rest at the origin, O , moves along the x -axis. The graph shows the acceleration, a , of the particle as a function of time t for $0 \leq t \leq 5$.



- Ⓐ At what time is the velocity of the particle a maximum?

- Ⓑ At what time is the particle furthest from the origin, O ? Give brief reasons for your answer.

- 8 A particle is initially at rest at the origin. Its acceleration, $\ddot{x}$ or $\frac{d^2x}{dt^2}$, as a function of time, t , is given by $\ddot{x} = 2\sin 4t$.

- Ⓐ Show that the velocity, $\dot{x}$ or $\frac{dx}{dt}$, of the particle is given by $\dot{x} = -\frac{1}{2}\cos 4t + \frac{1}{2}$.

HINT

Be careful of your graph transformations! Plot some key points so your graph is in the right place. t in radians!

- Ⓑ On a separate sheet of paper, sketch the graph of the velocity against time for $0 \leq t \leq 2$.

- Ⓒ Hence find the exact value of the time at which the particle first comes to rest after $t = 0$.

- Ⓓ Find the distance travelled by the particle between $t = 0$ and the time found in Part (c), correct to 3 decimal places.

- 9 The velocity of a body in metres per second is given by $v(t) = 2t - \frac{4}{\pi}\sin\frac{\pi t}{2}$. The position of the body is given by $x(t) = \int v(t).dt = \int \left(2t - \frac{4}{\pi}\sin\frac{\pi t}{2}\right).dt$.

- Ⓐ Find **an** expression for the position at time t , $x(t)$.

SLIP

Antidifferentiate and substitute carefully; there are many places you can slip. Also think about the difference between position, displacement and distance. A sketch graph on the calculator could help.

- Ⓑ The position initially ($t = 0$) is at the origin ($x = 0$). Use substitution to find the constant of integration, C .

- Ⓒ Hence find **the** expression, $x(t)$, for the position at time t .

- Ⓓ Find the exact position after 3 seconds ($t = 3$).

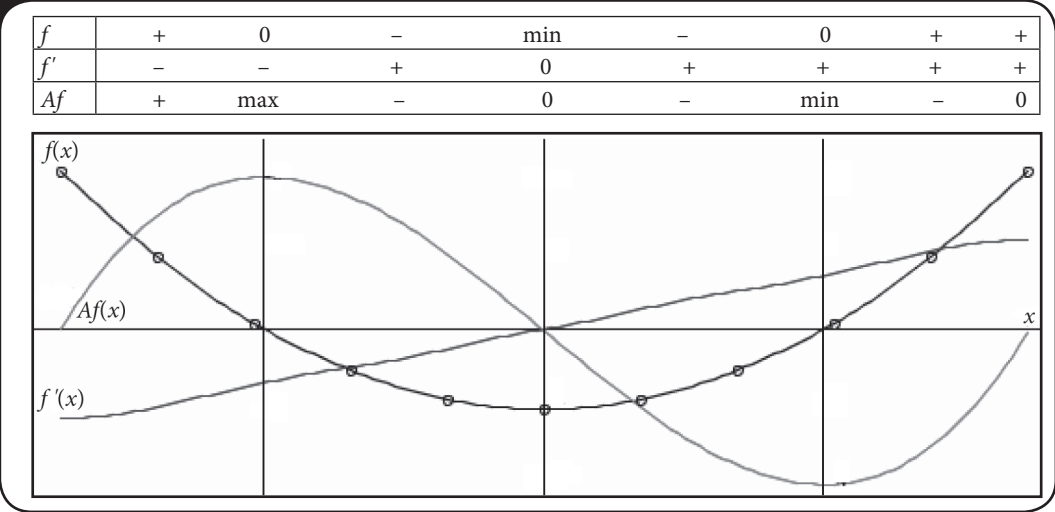
- Ⓔ Hence find the distance travelled by the body in the first 3 seconds.

9 – Integrals and graphs

It's OK to use a graphical calculator for learning, but sometimes you can't use it to answer a question. Make sure you have learned and understood how to find the relationship between the equation of the curve, the gradient of the curve, the area under the curve and the signed area under the curve – then you won't go far wrong. With these tools you should be able to answer questions when you haven't been given the equations.



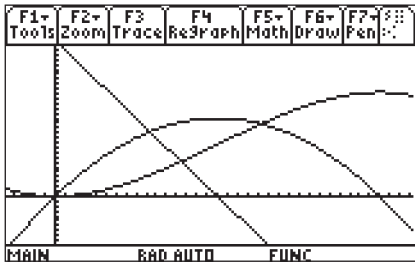
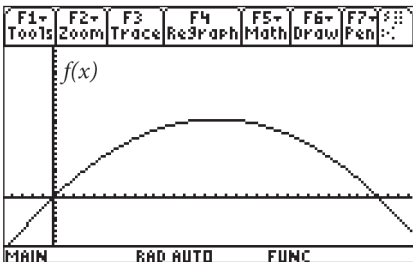
THINGS TO KNOW



HINT

You can't say that $f(x)$ is a quadratic and so the integral is a cubic because you haven't been given the equation. Better to look at the way the area under $y = f(x)$ changes.

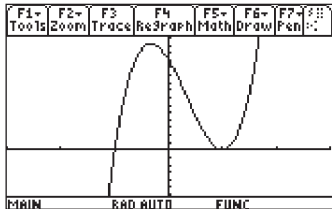
- 1 The graph of $y = f(x)$ is shown. The two other graphs drawn are $y = f'(x)$, the gradient function and $y = g(x)$ where $g(x) = \int f(x).dx$. Which of the two other graphs, the curve and the line, could represent $y = g(x)$? Explain your answer.



- 2 A Find the function f such that $f'(x) = 3x^2 - 3$ and $f(0) = 2$.

- B On a separate sheet of paper, sketch the graph of f , specifying the nature and coordinates of the turning points and the intercepts with the axes.

- 3 For the graph of $y = f(x)$ shown, the x -intercepts are at $(1, 0)$ and $(-1, 0)$ and the y -intercept is at $(0, 12)$. There is a minimum at $(1, 0)$ and a maximum at $(-\frac{1}{3}, 14\frac{2}{9})$. On the same axes, draw a rough sketch of the graph of $y = h(x)$ where $h'(x) = f(x)$ by completing the following table.



$h(x)$ increasing, decreasing, stationary	$h'(x) = f(x)$	Range of values of x
	positive	
	negative	
	zero	

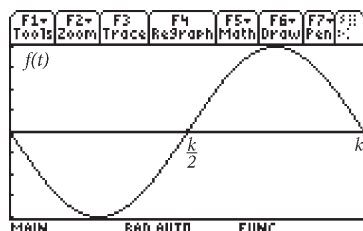
HINT

Getting a graph from its derivative is best done using tables like these. You can't use your calculator and you can't guess what the function is!



- 4 The graph of $y = f(t)$ is shown where k is a positive real number.

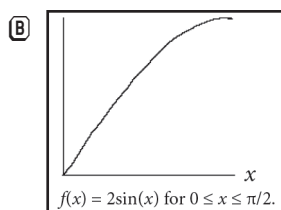
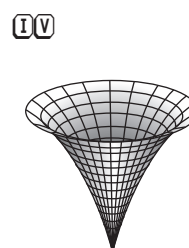
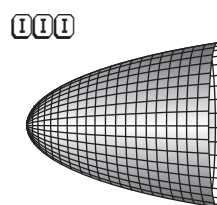
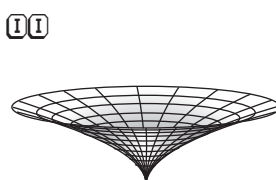
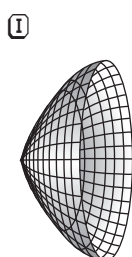
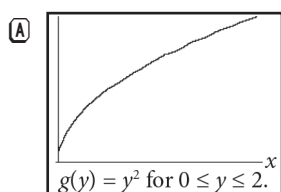
For $g: [0, k] \rightarrow \mathbb{R}$ where $g(x) = \int_0^x f(t).dt$, where is $g(x) < 0$?



THINGS TO KNOW

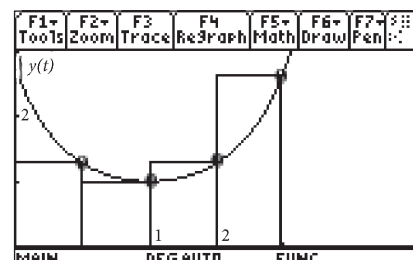
How do the graphs of the function and its integral relate? Check the summary above. Also remember the interval notation: $[$ and $]$ means include the end points, $($ and $)$ means don't include the end points; e.g. $[0, k]$ means $0 \leq t \leq k$ and $(0, k)$ means $0 < t < k$.

- 5 Three-dimensional solids are formed by rotating an area around an axis. Match the solid with the appropriate graph and state which axis is used to get each shape:



- 6 The following table of values of velocity $v \text{ ms}^{-1}$ at time $t \text{ s}$ have been plotted on a graph.

t	0	0.5	1	1.5	2
v	2.7	1.3	1	1.3	2.7



In order to estimate $\int_0^2 v(t).dt$, rectangles have then been drawn at intervals under the graph as shown.

- A What is the width of each rectangle? _____

- B List the heights of each of the four rectangles. _____

- C Find an estimate of the area under the curve between $t = 0$ and $t = 2$.

HINT

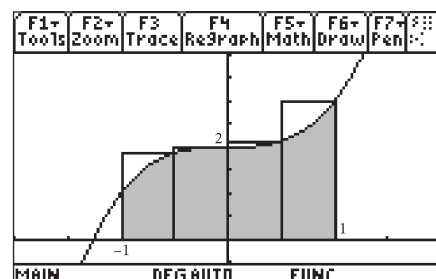
Use the table of values, don't estimate from the graph!

- 7 The graph shown has the rule $y = x^3 + 2$. Rectangles of width 0.5 units have been drawn on the graph so as to estimate the area under the graph between $x = -1$ and $x = 1$.

- A List the heights of each of the four rectangles.

- B Find an estimate of the area under the curve between $x = -1$ and $x = 1$.

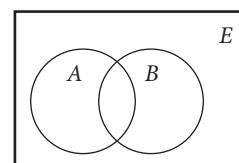
- C Find the exact value of $\int_{-1}^1 (x^3 + 2).dx$.



10 – Probability and events

You need to understand the basic principles of probability as well as be able to complete problems involving the concepts of dependent, independent, mutually exclusive, and conditional probability.

Sometimes Venn or tree diagrams can be helpful.



THINGS TO KNOW

Mutually exclusive events A and B do not intersect and so:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B).$$

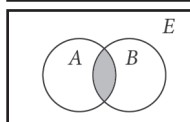
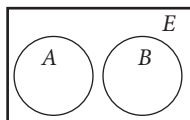
Non-mutually exclusive events A and B intersect and so:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B).$$

If the event B is a subset of event A then $\Pr(A \cap B) = \Pr(B)$.

If events A and B are independent then

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B).$$



If events A and B are dependent then

$$\Pr(A \cap B) = \Pr(A|B) \times \Pr(B) \text{ and}$$

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B|A).$$

So $\Pr(B|A) = \frac{\Pr(A \cap B)}{\Pr(A)}$ where $\Pr(B|A)$ denotes the probability the event B occurs given or knowing that event A has occurred.

Remember that the probability of 'not B ' is

$$\Pr(B') = 1 - \Pr(B).$$

- ① Copy and complete the following statements with the words **independent**, **mutually exclusive**, **dependent** or **not mutually exclusive** to make each statement true:

Ⓐ $\Pr(A \cap B) = \Pr(A) \times \Pr(B)$ if A and B are _____.

Ⓑ $\Pr(B|A) = \Pr(B)$ if A and B are _____.

Ⓒ $\Pr(A \cap B) = 0$ if A and B are _____.

Ⓓ $\Pr(A \cup B) = \Pr(A) + \Pr(B)$ if A and B are _____.

Ⓔ $\Pr(A \cap B') = \Pr(A) \cdot \Pr(A'|B)$ if A and B are _____.

Ⓕ $\Pr(A \cap B') = \Pr(A) - \Pr(A \cap B)$ if A and B are _____.

- ② Find $\Pr(B)$ if:

Ⓐ A and B are two independent events with $\Pr(A) = 0.3$ and $\Pr(A \cap B) = 0.06$.

Ⓑ Events A and B are such that $\Pr(A \cup B) = 0.8$, $\Pr(A \cap B) = 0.2$ and $\Pr(A) = 0.55$.

- ③ Find $\Pr(A \cup B)$ if A and B are events such that $\Pr(A) = 0.55$, $\Pr(A \cap B) = 0.25$ and $\Pr(B) = 0.4$.

- ④ If A and B are events such that $\Pr(A) = \frac{3}{4}$, $\Pr(B') = \frac{1}{4}$ and $\Pr(B' \cap A) = \frac{1}{8}$, find.

Ⓐ $\Pr(B)$. _____

Ⓑ $\Pr(A \cap B)$. _____

Ⓒ $\Pr(A \cup B)$. _____

- ⑤ Find $\Pr(A \cap B)$ if A and B are events such that:

Ⓐ $\Pr(A) = 0.5$, $\Pr(B) = 0.4$ and $\Pr(A \cup B) = 0.7$. _____

HINT

Venn diagrams will help!

HINT

Use Venn diagrams and the definitions listed above.



- 6 A and B are mutually exclusive. _____
- 7 $Pr(B) = 0.5$, $Pr(A|B) = 0.4$. _____
- 8 Find $Pr(B|A)$ if A and B are events such that $Pr(A) = 0.6$, $Pr(A \cap B) = 0.4$. _____
- 9 Find the greatest value $Pr(A|B)$ can have if $Pr(A) = 0.5$ and $Pr(B) = 0.6$. For this case, show the events A and B on a Venn diagram and mark in the probabilities.

- 10 Two dice, one white and one black, are placed in a box. One die is selected at random and tossed. What is the probability that:
- A the result is a 6 on the white die? _____
- B the result is a 6? _____
- C the result is not a 6? _____
- D given that the selected die is white, the result is a 6? _____
- E if the result is a 6, it is on the white die? _____
- F if the result is not a 6 then it is on the black die? _____
- 11 There are 150 jelly beans in a jar: 50 red, 50 black and 50 yellow. Two jelly beans are selected at random. Use a tree diagram to find the probability that:
- A the two selected jelly beans will be different colours.

- B at least one jelly bean is **not** black.

- 12 The probability that it rains on a particular day of my holidays is dependent on whether or not it has rained on the previous day. If it has rained the day before, the probability of rain is 0.7. If it has not rained on the previous day then the probability of rain is 0.2. If it rained on Saturday, what is the probability that it will **not** rain on the following Monday?

- 13 A fisherman has two favourite fishing spots on the river: at the weir and at the pond. Where he fishes depends on where he fished last time. If he fished at the weir last weekend then the probability that he will fish at the weir the next weekend is 0.6. If he fished at the pond last weekend then the probability that he will fish at the pond the next weekend is 0.3. Last weekend he fished at the weir. What is the probability that:
- A he will fish at the weir on the next three weekends? _____
- B he will fish at the pond for the first time three weekends from now? _____
- C he will fish at the weir on only one of the next three weekends? _____

HINT

A tree diagram will help!

HINT

You are asked to use a tree diagram so make sure you show it.

HINTYour tree could show events B and B' only rather than all the colours. In that case $Pr(\text{at least one not black})$ can be worked out using $1 - Pr(\dots)$ if you wish.

11 – Tables and counting

As well as Venn diagrams and tree diagrams, you can use other ways of counting possibilities. These include two-way tables (Karnaugh diagrams) and permutation techniques. Of course you don't have to use either unless the examiner tells you to!



- ① The following two-way table shows how people voted in a local election.

	Male voters	Female voters	Total
Industry Party	1237	2369	
Commercial Party	2549	1395	

HINT

You should have a basic understanding of reading a two-way table. Also be careful you don't transpose the columns, and keep the question parts clearly separate and well labelled.

One female voter was selected at random from all of the female voters. What is the exact value of the probability that she voted for the Industry Party?

HINT

Notice the words 'exact value'.

- ② Each of the ten people at a table for a fundraising dinner has a place-card with their name on it. The men's place-cards are blue and are even-numbered (2, 4, 6, 8, 10) and the women's are red and odd-numbered (1, 3, 5, 7, 9).

- A Complete the two-way table.

One person is selected at random.

- B What is the probability that:

	Red card	Blue card	Total
Number of cards divisible by 4			
Number of cards not divisible by 4			

I they have a red card? _____

II they have a card with a number divisible by 4? _____

III either they have a red card or a card with a number divisible by 4 or both? _____

IV they have a red card and a card with a number divisible by 4? _____

- ③ Alcohol tests are not always accurate. An alcohol test was administered to 400 people. The results were:

- ☞ 100 people had drunk some alcohol – of these, the test indicated that 80 had drunk some alcohol
- ☞ 300 people had not drunk alcohol – of these, the test indicated that 40 had not drunk some alcohol.

- A Copy and complete the table using the information above.

	Number the test indicated had drunk some alcohol	Number the test indicated had not drunk alcohol	Total
People who had drunk some alcohol			100
People who had not drunk alcohol			300

- B For how many of the people tested was the alcohol test accurate?

- C For what percentage of the people tested was the alcohol test accurate?

- D For what proportion of the people tested was the alcohol test **not** accurate?

TRAP

This question asks for the number of people, not the fraction, in part (b) and percentage, proportion and probability in parts (c), (d) and (e). Make sure you use your table for your answer so you get consequential marks. Converting a fraction into a percentage is a skill you need to have.



Ⓔ What is the probability that a randomly selected person tested negative when in fact they had drunk alcohol?

- Ⓖ The number, X , of puppies produced by bitches of a particular breed is a random variable with probability distribution as shown in the table. Find the probability that two bitches selected at random will have the same number of puppies.

x	0	1	2	3	4
$Pr(X = x)$	0.3	0.2	0.25	0.15	0.1

THINGS TO KNOW

You need to revise the method you used for 'permutations' or 'arrangements'.

Permutations are arrangements so order is important. To find the number of arrangements of n things, r at a time:

$${}^nP_r = n(n-1)(n-2)(n-3) \dots (n-r+2)(n-r+1)$$

$$\text{Or } {}^nP_r = \frac{n!}{(n-r)!}$$

The number of arrangements of n things taken n at a time is

${}^nP_n = n(n-1)(n-2)(n-3) \dots 3 \times 2 \times 1 = n!$ where $n!$ is called ' n factorial' and $0! = 1$. Also, the number of arrangements of n things comprising p of one type, q of another type, r of another type, etc. is: $\frac{n!}{p!q!r! \dots}$

Combinations are selections so the order is not important.

To find the number of selections from n things taken r at a time = the number of ways of arranging n things taken r at a time $\div$ the number of arrangements within each selection:

$${}^nC_r \text{ or } \binom{n}{r} = \frac{{}^nP_r}{r!} = \frac{n!}{(n-r)!r!} = \frac{n(n-1)(n-2) \dots (n-r+1)}{r!}$$

Also, the number of selections from n things r at a time = the number of selections from n things $(n-r)$ at a time:

$${}^nC_{n-r} = {}^nC_r \text{ or } \binom{n}{n-r} = \binom{n}{r}$$

When there are restrictions placed on arrangements or selections they must be dealt with first.

- Ⓗ In the word game called Scrabble, I drew 7 different letters from a bag and placed them in a row on a rack in front of me.

Ⓐ In how many ways can the letters be arranged? _____

If I drew 4 consonants and 3 vowels, what is the probability that they will be arranged with:

Ⓑ the consonants and vowels alternating? _____

Ⓒ the consonants together? _____

- Ⓘ Ⓐ In a beer-tasting competition, Ray was asked to arrange six different beers in order of quality. He could not distinguish between any of the beers. If the beers comprised three ales, one pilsener, one lager and one light, in how many different ways can he arrange the beers in a row? Show working to justify your answer.

Ⓑ In how many of Ray's arrangements are the ales grouped together?

Ⓒ Find the probability of Ray placing the three ales together.

Ray's mate, Bob, chooses two beers at random from the row.

Ⓓ What is the probability that he chooses two ales?

Ⓔ What is the probability that he chooses an ale first and a light second?

HINT

The need to 'justify your working' means you need to show how you got the answer by counting and multiplying (not adding!) or listing the arrangements. The tree diagram may not be the best way to do this question.

Using $\frac{\text{number of favourable arrangements}}{\text{total number of arrangements}}$ may be easier but be careful with your fraction work.

Maybe you should use your calculator to check your calculations. Part (e) has dependent events so be careful.

12 – Discrete probability questions

Some students will call these questions 'Binomial Probability' questions and will use a formula. Some will work them out by first principles or with a tree diagram. You should try these questions using the method that your teacher suggests.



- ① The amount of mineral water in a bottle is advertised to be 600 mL. The probability that there is less water than the labelled amount is 0.12. For 5 randomly selected bottles of water, find the probability, correct to three decimal places that:

A 2 contain less than the labelled amount.

B at least 2 contain less than the labelled amount.

TRAP

Remember that the probability of 'at least 2' is not $1 - \text{probability of 'at most 2'}$.

Also, $Pr(x \geq 2) \neq 1 - Pr(x \leq 2)$.

- ② If the probability of selecting a boy over 2 metres tall in a town is $\frac{20}{81}$,

A what is the probability, correct to three decimal places, that a sample of 8 has exactly 2 boys over 2 metres tall?

B what is the smallest sample size that should be taken so that the probability of selecting no boys over 2 metres tall is at most 0.09?

TRAP

For the probability that needs to be correct to 3 decimal places make sure you show all working and are not careless with the rounding. Notice the words 'smallest sample', 'no boys' and 'at most'. The 'smallest value' can be found by trial and error or solution of an equation.

- ③ A selection of balls is made from a box containing white and black balls. Each ball is replaced before the next is taken.

A If the mean number of black ball selected is 1.6 with a variance of $\frac{24}{25}$, find:

① the probability of selecting a black ball on any one trial.

①① the number of balls selected.

B If the box is known to contain 15 balls, how many black balls and white balls were there in the box?

THINGS TO KNOW

You need to know that for a binomial experiment the mean value of the variable is given by $\mu = np$ and the variance by $\sigma^2 = npq$. Then of course you can use simultaneous equations.

- ④ A machine making car parts is checked every hour. A random sample of 10 is taken as they leave the machine and if fewer than 2 defective parts are found in the sample then the machine is allowed to keep going. But if there are 2 or more defective parts then the machine is stopped and adjusted. What is the probability, correct to three decimal places, that the machine is allowed to keep going even though 8% of the parts being produced by the machine are defective?



12 – Discrete probability questions (cont'd)

TRAP

This is not a conditional probability question! And a tree diagram helps.

HINT

You need conditional for this one!

HINT

If you get a probability greater than 1 then you must have made a mistake! Don't just leave the answer there without commenting on the appropriateness of the answer.

TRAP

Read the question carefully. The proportion is not a number. Also notice that the selection is with replacement. Watch out for the domain of the graph. The second part needs calculus to find the turning point. Which one?

TRAP

Remember that $\Pr(x \geq 7) \neq 1 - \Pr(x \leq 7)$. And watch the significant figures; don't round off on the way to the answer.

- _____
- _____
- B** Two machines, a Smith brand and a Wesson brand, are used to make car parts. The Smith makes 20% of the parts and the Wesson makes 80%. If 15% of the output of the Smith machine and 5% of the output of the Wesson machine is defective, find the probability that a randomly selected item is defective.
- _____
- _____

- 5** Amanda works as a scientist and experience has shown her that 75% of her applications for funding are successful. What is the probability, correct to 4 decimal places, that of the next 15 applications, at least 12 will be successful?
- _____
- _____

- 6** In a game of chance, whether or not a single game is won is independent of the result of any other game. The probability of winning a single game is 0.12. Gen played a series of n games. Find the smallest number of games she should play so that the probability of her winning at least one game is more than 0.95.
- _____
- _____

- 7** Three dice are taken from a box of 100 dice, some of which are 'loaded'. The proportion of loaded dice in the box is denoted by p . If a selection of three dice is made **with replacement** (that is, a die is selected, it is noted whether the die is loaded or not, and then the die is replaced before the next die is selected):

- A** write an expression for the probability P that exactly one of the three dice selected is loaded.
- _____

- B** on a separate sheet of paper, draw a graph of P against p .

- C** use Calculus to find the exact value of p for which this probability will be a maximum.
- _____
- _____

- 8** **A** A multiple choice question has 5 possible responses of which only one is correct. Henry didn't know the answer and so decided to choose a response randomly. What is the probability that he chose the correct response?
- _____
- _____

- B** Section A of the exam paper included 10 such multiple choice questions. What is the probability, correct to 3 significant figures, that if Henry picked random answers to all the questions, he chose:

I 7 correct responses? _____

II 7 or more correct responses? _____

13 – Probability and selections

Some students will call these questions *Hypergeometric Probability* questions, and will use a formula. Some will work them out by first principles or with a tree diagram. You should try these questions using the method that your teacher suggests. Notice that there are lots of traps in these questions so read them carefully.



- ① Of the 20 students in the Year 12 Mathematics class, only 2 do not have a computer at home. If 5 students are selected at random, what is the probability, correct to 4 decimal places, that:

Ⓐ only 3 of them have a computer at home?

Ⓑ no more than one does not have a computer?

TRAP

Watch the sneaky wording!

- ② In a draw of raffle tickets numbered from 1 to 99, two lucky numbers are selected at the same time at random from a hat.

Ⓐ In how many ways can the two lucky numbers be selected?

HINT

'At the same time' is like 'without replacement'.

Ⓑ John bought two tickets in this raffle. What is the probability that at least one of his two numbers will be selected?

TRAP

Read the question carefully: 'pairs of socks' and 'individual socks'. This is the sort of trap you should notice during reading time. And 'exact value' means no decimals.

- ③ I always buy the same socks and only ever buy either white or black. At the moment there are 6 pairs of black socks and 4 pairs of white socks in my drawer. When I got up this morning in the dark I grabbed two individual socks. What is the exact value of the probability that they matched?
- _____

- ④ Packets of printing paper come in reams (500 sheets) which are then packed in boxes. A number of boxes of printing paper was stained by water in a recent flood. If a random sample of 3 reams of paper was selected from a box of 12 reams in which 4 reams were stained, what is the probability that at least one ream was stained?
- _____

HINT

Note the difference between 'only one stained' ($= 1$) and 'at least one stained' (≥ 1).

- ⑤ The local school decided to hold a lucky draw with 450 tickets being sold. 150 tickets were pink, 150 were yellow and 150 were blue and there were to be three lucky tickets drawn. Winning tickets were **not** replaced before the next ticket was drawn. Find the probability, correct to 3 significant figures, that:

Ⓐ each of the winning tickets is pink.

Ⓑ all the tickets are of different colours.

Ⓒ at least one of the winning tickets is not blue.

HINT

Notice that the tickets are not replaced before the next draw, so the probability changes for each draw. Also make sure you show all of your working: sometimes an answer is approximately the same as the correct answer and the examiner needs to know how you got your answer. Read the question carefully, like 'not blue'.



13 – Probability and selections (cont'd)

- 6 A box contains 3 white balls and k black balls. When two balls are drawn at the same time at random from the box, the probability that both balls are the same colour is $\frac{1}{2}$. But when one ball is replaced before the second is taken, the probability that both balls are the same colour is $\frac{5}{8}$. What are the possible numbers of black balls in the box; that is, what are all the possible values of k ?

- 7 A florist has noticed that 30% of customers select roses. What is the probability that of the next 20 customers, at least 18 will select roses?

- 8 There are 150 children's blocks in a box: 50 red, 50 green and 50 yellow. Three blocks are selected at random without replacement.

- A What is the probability that each of the three blocks is red?

- B What is the probability that there are no blocks that are not red?

- C What is the probability that there is only one red block?

- 9 A magician shuffles 10 cards that are either red or black and asks you to take two. If the first card is red then the probability that the second card is red is $\frac{2}{9}$, while if the first card is black then the probability that the second card is red is $\frac{1}{3}$.

- A In the original pack of 10, how many cards were red and how many were black? _____

- B Was the first card replaced before the second card was taken? _____

- C What is the probability of 2 black cards being selected? _____

- D What is the probability of 1 red and 1 black card being selected? _____

- E If the first card is black, what is the probability that the second card is black? _____

- F If the second card is red, what is the probability that the first card is red? _____

- 10 A card game was being played with an ordinary pack of 52 cards in 4 suits. The first card dealt was the five of spades. Put the probabilities of the following cards being the second card dealt in order from smallest to largest: a spade; a heart; a five; the five of spades; not the five of spades.

TRAP

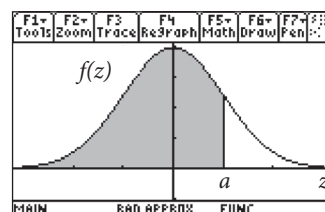
Is this really a selection question? What do you do if you find a probability greater than 1? You comment on it at least!

HINT

When the tree diagram is too big you need to use a different method because selection without replacement can be confusing. When you are dealing with large numbers, incorrect decimal answers can be very close to correct decimal answers so it is essential that you show all your working.

14 – Normal distribution

Make sure you understand the standard normal distribution $f(z) = \sigma f(x)$ where $z = \frac{x - \mu}{\sigma}$, sometimes denoted by $N(0, 1)$. The graph of $f(z)$ is symmetrical about $z = 0$ and is shown here with $\Pr(Z \leq a)$ being area between the curve of $f(z)$ and the z -axis from $-\infty$ to a .



THINGS TO KNOW

For a normal random variable, X , with mean μ and standard deviation σ , the probability density function is

$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$ for $z \in R$. As a general rule, you should show all your working with any diagrams, graphs and equations that you used. You should be able to use the normal distribution tables or a calculator to find probabilities and if you use the calculator, make sure that you indicate the parameters (μ and σ) that you used.

- ① What are the values of the mode, mean and median of a standard normal distribution?

HINT

Know the properties of the standard normal distribution.

- ② For a normal distribution with mean 10.8 and standard deviation 1.6, copy and complete the following with exact values:

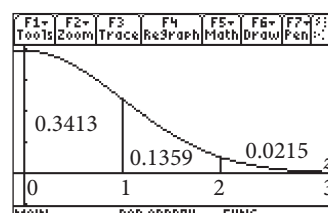
$$Pr(X > 8) = Pr(Z > \underline{\hspace{1cm}}) = Pr(Z \leq \underline{\hspace{1cm}}).$$

- 3** A student completing a probability question in the exam wrote $Pr(X \leq 36) = Pr(Z \leq 1.439)$. What was her answer correct to 4 decimal places?

HINT

It is always useful to draw a diagram. Can you use the tables or calculator when you don't know the mean μ and standard deviation σ for x ?

- 4** The right-hand half of the graph of $f(z)$ against z is shown with the areas of the three sections labelled. Use this information to find:



- Ⓐ $Pr(0 \leq Z \leq 2)$. _____

- ⓑ $Pr(Z \leq 2)$.

- © $Pr(Z > 2)$.

- ⑩ $Pr(Z \leq -2)$.

- Ⓔ $Pr(Z > -1)$.

- Ⓕ $Pr(-3 \leq Z \leq -1)$.

- Ⓔ the whole number value of Z below which 99.87% of values lie.

- ⑤ For a normal random variable, it is said that the values between which X will **almost certainly** lie are given by $\mu - 3\sigma \leq x \leq \mu + 3\sigma$. For a normal random variable with mean 12 and standard deviation 4:

- (A) over what range will x almost certainly lie? _____

- ⓑ what is the probability that x will lie within that range? _____



14 – Normal distribution (cont'd)

- 6 Boxes of sultanas are labelled as containing 250 g. In fact, the weight of the contents is normally distributed with mean 250.5 g and standard deviation 0.75 g. Find the proportion of boxes on the supermarket shelves that contain less than the labelled weight of sultanas.

HINT

Read the question carefully: what does it mean to be 'less than the labelled weight'? Proportion here means probability.

- 7 In a school test, the marks were normally distributed with a mean of 50 and a standard deviation of 10. If only 15% achieved a B grade or better, what is the minimum whole number mark a student needs to get a B?

HINT

Working backwards from probability to the value of Z and then to the value of X requires care as well as an understanding of the standard normal graph and of $z = \frac{x - \mu}{\sigma}$. You must show all your working, particularly if you are working backwards. Give your answer with the correct accuracy.

- 8 A university survey was carried out to find information on the heights of students entering a first year course. Over the long term it was found that the heights were normally distributed with mean 178 cm and standard deviation 12.8 cm. What proportion of first year students are between 159 cm and 192 cm tall?

TRAP

Even though data is being collected (statistics), the question does say that the heights are 'normally distributed' so you will need to use that. Also, real data doesn't always give 'nice' numbers. Watch round-off!

- 9 Babies born at a particular hospital have a mean length 50 cm with standard deviation 6 cm. If the distribution of babies' lengths is normal and it is known that 15% of babies are less than k cm long, find the value of k correct to 1 decimal place.

TRAP

Be careful when there are three variables, x , z and k . Notice that k is a value of x and z is associated with the standard normal.

- 10 The amount (in dollars) paid out by a particular insurance company on claims in any one year is normally distributed with mean \$4900 and standard deviation \$1150.

- A Find the probability that a claim selected at random:

i will be greater than or equal to \$5100.

ii will be less than \$500.

- B Mike processed a sample of 12 claims last Tuesday morning. What is the probability, correct to two decimal places, that only 4 of them will be for a more than average claim?

TRAP

Tricky – binomial, and what is the probability of an 'above average claim'?

15 – Coordinate geometry

As with all maths questions you should make sure that you know all your rules and definitions: angle between two lines, gradient, distance between two points, $y = mx + c$, etc. A diagram will always help you and, above all, don't let slips in your algebra let you down; there are not always just numbers in the question.



THINGS TO KNOW

For two points $A(x_1, y_1)$ and $B(x_2, y_2)$:

$$\text{Distance } AB = \sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}$$

Midpoint of line AB :

$$M(x, y) = M\left(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2}\right)$$

Gradient of AB :

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan \theta$$

where θ is the angle AB makes with the positive x -direction.

For two lines with gradients m_1 and m_2 :

- The lines are parallel if $m_1 = m_2$
- The lines are perpendicular if $m_2 = \frac{-1}{m_1}$ or $m_1 \times m_2 = -1$
- The angle between the two lines ϕ can be found using:

$$\tan \phi = \pm \frac{m_2 - m_1}{1 + m_2 m_1}$$

(A negative value for $\tan$ means the angle is in the 2nd or 4th quadrant.)

① Is the gradient of the line joining the points (1, 3) and (3, 1) positive or negative? _____

② For the points $A(2, -1)$ and $B(-3, 4)$, find:

Ⓐ the distance from A to B .

Ⓑ the midpoint, M , of the line AB .

Ⓒ the gradient of the line AB .

Ⓓ the gradient of the line perpendicular to the line AB .

Ⓔ the equation of the line through M perpendicular to the line AB .

③ Which of the following statements are not true?

Ⓐ The line with equation $7x - 4y = 0$ passes through the point (4, 7).

Ⓑ The point (3, 5) lies in the region with inequation $7x - 4y \geq 0$.

Ⓒ The line with equation $3x + 5y = 0$ has a positive gradient.

Ⓓ The lines with equations $7x - 4y = 0$ and $3x + 5y = 0$ meet at the point (0, 0).

Ⓔ For the line with equation $3x + 5y = 0$, y increases as x increases.

HINT

Be careful of the signs!

HINT

Test systematically!



- 4 The equation of the normal to the line with equation $y = 3x - 1$ at the point $(1, 2)$ is given by $\frac{y-2}{x-1} = -\frac{1}{3}$. Express this equation with y the subject.

TRAP

Read the question carefully and highlight what is being asked.

- 5 The angle between the lines with equations $6x + 2y = 3$ and $2\sqrt{3}x - 2y = 7$ is given by $\tan \alpha = \frac{-3 - \sqrt{3}}{1 - 3\sqrt{3}}$. Find the value of α in degrees correct to 2 decimal places.

TRAP

Degree mode not radian mode!
Acute or obtuse angle?

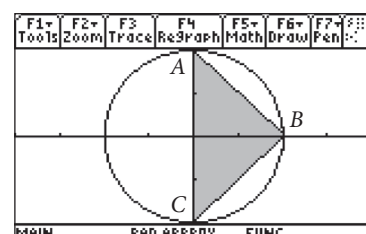
- 6 Find the area of the space enclosed by the lines with equations:
 $x + y \leq \frac{1}{4}$; $x + y \geq -\frac{1}{4}$; $y - x \geq -\frac{1}{4}$; $y \leq x + \frac{1}{4}$

HINT

Draw a diagram! Be careful if the scale on the graphical calculator is misleading.

- 7 The diagram shows the graph of $x^2 + y^2 = 4$ with $A(0, 2)$, $B(2, 0)$ and $C(0, -2)$.

- A Find the lengths of AB and BC .

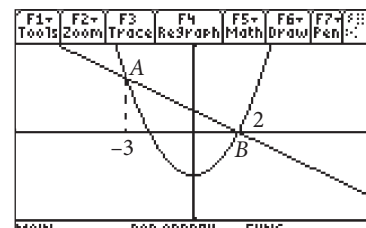


- B Show that $\angle ABC$ is 90° .

- C Hence find the area of the triangle ABC .

- 8 The diagram shows the graphs of $y = x^2 - 4$ and a line joining point A with x -coordinate -3 and B with x -coordinate 2 .

- A Find the coordinates of A and B .



- B Find the equation of the line joining A and B .

- 9 On the diagram, the equation of the line $\overline{OP}$ is $x - 2y = 0$ and the equation of the line $\overline{AB}$ is $5x - 2y = 20$. C is the point $(9, 2\frac{1}{2})$ and D is the point $(8, 0)$.

- A Find the coordinates of point B .

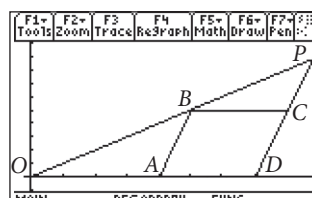
- B Find the exact distance between points C and D .

- C Which lines on the diagram are parallel and why?

- 10 Find the coordinates of the focus of the parabola with equation $y^2 = 20(x - 3)$.

SLIP

Be careful of algebraic errors when solving simultaneous equations – you want the exact distance and your reasons should be argued logically. Learn the distance formula if you don't already know it.

**THINGS TO KNOW****Conic sections**

For the equation of the parabola $y^2 = 4ax$, the focus is $(a, 0)$, the directrix is $x = -a$, the axis of symmetry is $y = 0$ and the vertex is $(0, 0)$.

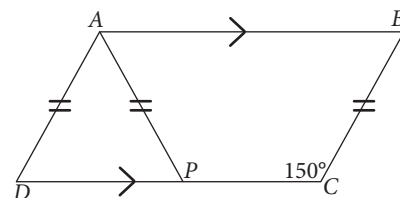
16 – Geometry and trigonometry

It's a good idea to practise drawing diagrams neatly and accurately. It's also important to know your geometry facts so that you can make a logical argument. Remember to write in the reasons for each step beside it.



① In the diagram shown:

- Ⓐ Name the parallelogram. _____
- Ⓑ Name the isosceles triangle. _____
- Ⓒ Name an angle equal to $\angle ADP$. _____
- Ⓓ Use a logical argument to explain why $\angle ADP = 30^\circ$.



THINGS TO KNOW

Note: make a logical argument; write in the reasons beside each step.

② The lengths of the sides of a triangle are 6 cm, 7 cm and 10 cm.

- Ⓐ Find the size of the angle opposite the longest side.

- Ⓑ Find the area of the triangle.

③ For a triangle with side lengths of 6 cm, 7 cm and 12 cm, find in degrees correct to 1 decimal place:

- Ⓐ the angle opposite the longest side.

- Ⓑ the smallest angle.

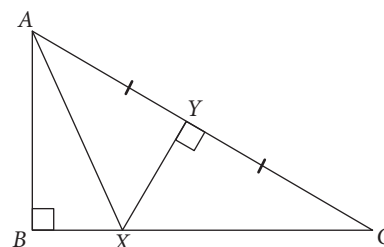
④ The diagram shows a right-angled triangle ABC with $\angle ABC$ a right angle. The point Y is the midpoint of $\overline{AC}$, and X is the point where the perpendicular to $\overline{AC}$ at Y meets $\overline{BC}$.

- Ⓐ Why are $\triangle AXY$ and $\triangle CXY$ (congruent) identical?

- Ⓑ Find the size of $\angle AXB$, $\angle AXY$ and $\angle YXC$ in degrees if it is known that AX bisects $\angle BAC$.

- Ⓒ Hence find the size of $\angle XCY$ in degrees.

- Ⓓ What is the exact ratio $YX:AC$?



TRAP

Read the question carefully and watch out for negative cosine. Use the correct mode for the calculator!

HINT

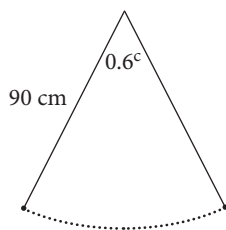
Read the question: label your diagram so you can see which is the angle opposite the longest side and which is the smallest angle. What does a negative cosine mean? Careful of round-off on the way to the answer.

THINGS TO KNOW

Exact ratio so write as $a:b$ or $\frac{a}{b}$.



- 5 A pendulum of length 90 cm swings through an arc of 0.6 radians. Find:
- A the length of the arc of a pendulum.



- B the straight line distance between the extremes of the swings to 4 significant figures.
- C the exact value of the area of the sector swept out by the pendulum.

TRAP

Show the correct substitution in the formula and don't make errors like:

- $0.6\pi \neq 108^\circ$ and θ should not be written as 34°
- signs in the cosine rule can be tricky
- the $\sqrt{\quad}$ sign is needed in your calculation
- know the difference between a sector and a segment.

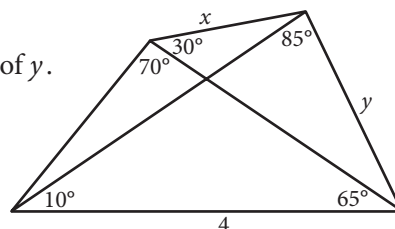
It should be such a simple question, but it has lots of traps.

- 6 Here is a wordy problem:
'A hiking group wants to walk from their overnight camp at point A to a hut at point B 12 km away (as the crow flies) on a bearing of 105° . They can't walk straight there because there is a high mountain in their way. Instead they walk first to point C which is 4 km on a bearing of 120° , then to point D which is 9 km from point C on a bearing of 60° . Find the bearing, x° , and distance, y km, to point B when they arrive at point D.
Draw the diagram and mark on it as x and y what you are asked to find.

HINT

The examiners say things like 'Those candidates who drew sketches were able to progress towards a solution.' So surely you can do these questions if you have a diagram. Be careful to think about whether it is 2-dimensional or 3.

- 7 A Use this diagram taken from a bearing problem to find the value of y .



- B Hence or otherwise find the value x correct to 1 decimal place.

HINT

Don't correct the number of decimals for the value of y before going on to get x .

- 8 The diagram below came from a 'wordy' problem. Use it to find each of the following correct to 1 decimal place.

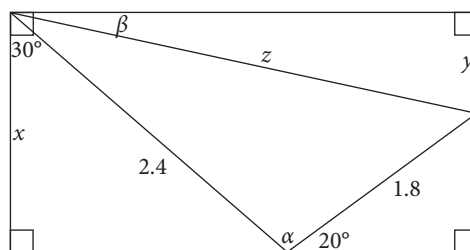
A x _____

B y _____

C α _____

D z _____

E β _____



TRAP

Consequential errors (errors following because of previous errors) can be a problem. Also make sure that you use the Cosine Rule correctly: $c^2 = a^2 + b^2 - 2ab\cos C$ and that you don't round too soon; wait until the end.

17 – Sequences and series

Number patterns are useful in many branches of mathematics. You need to know the rules for finding the general term or sum of a sequence of numbers. As usual, real life examples can have 'yucky' formulae but they are not necessarily difficult. If this topic is not in your syllabus this year, you can still complete most of the questions just because they give you good practice in your algebra and arithmetic.



THINGS TO KNOW

Arithmetic sequence: $t_n = a + (n-1)d = \ell$ where t_n is the n^{th} term, a is the first term, $d = t_2 - t_1 = t_3 - t_2 = \dots = t_n - t_{n-1}$ is the common difference and ℓ is the last or n^{th} term.

Arithmetic series: sum to n terms: $S_n = \frac{n}{2}(a + \ell)$ or $S_n = \frac{n}{2}[2a + (n-1)d]$.

Geometric sequence: $t(n) = ar^{n-1} = \ell$ where a is the first term and $r = \frac{t_2}{t_1} = \frac{t_3}{t_2} = \dots = \frac{t_n}{t_{n-1}}$ is the common ratio and ℓ is the last or n^{th} term.

Finite geometric series sum to n terms:

$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{r\ell - a}{r - 1}$ convenient if $|r| > 1$ or

$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a - r\ell}{1 - r}$ convenient if $|r| < 1$

Sum to infinity or limiting value: $S_\infty = \frac{a}{1 - r}$ where $|r| < 1$.

- ① To check that you have the basic skills, calculate the following.
 - Ⓐ Find the 13th term of the sequence {3, 9, 15, 21, ...}. _____
 - Ⓑ Find the sum of the multiples of 6 between 14 and 45. _____
 - Ⓒ Find ⑩ the tenth term and ⑩⑩ the number of terms in the sequence {2, 8, 32, 128, ..., 131 072}.

 - Ⓓ Find the series (sum of the sequence) $3 + \frac{3}{4} + \frac{3}{16} + \dots + \frac{3}{1024}$, correct to 3 decimal places.

 - Ⓔ Evaluate: $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$ _____
 - ② If for a geometric series the n th term is given by $t_n = 5 \times 0.9^{n-1} + \frac{1.2(1-0.9^{n-1})}{1-0.9}$, find t_6 correct to 3 decimal places.

 - ③ Why is $4\sin\theta + 4\sin^2\theta + 4\sin^3\theta + \dots$ a geometric series? Find the sum to infinity of this series.

 - ④ A long distance runner sets up a training regime for which she runs 5.3 km on the first day, 5.6 km on the second day, 5.9 km on the third day and so on. Find the distance she ran from Day 3 to Day 12 inclusive, correct to 1 decimal place.

- HINT**

Arithmetic sequence but what are the values of a , d and n ?
- ⑤ For the two series $60\,000 + 63\,000 + 66\,000 + \dots$ and $60\,000 + 63\,000 + 66\,150 + \dots$:
 - Ⓐ find the 5th term of each.

**HINT**

Arithmetic or geometric. Picking n can be tricky.

- ⓑ find the term at which the second sum *first* becomes greater than the first sum.

- ⓖ Eleanor and Lillian just started work in Bill and Son's office on the same annual salary, \$48 000 per year. Eleanor's contract gave her a salary increase of \$2000 per year in each of the next ten years. Lillian's contract gave her a salary increase of 4% per year in each of the next ten years.

- Ⓐ Who earned the larger salary in her second year?

- Ⓑ What was Eleanor's salary in her tenth year?

- Ⓒ What was Lillian's salary in her tenth year?

- Ⓓ By how much did the total of ten years' salaries for Lillian exceed that paid to Eleanor?

- ⓓ The balance owing after two years on a loan of \$5 million dollars at an annual interest rate of 9% per annum is given by $B = (5 \times 10^6 \times (1.09)^2 - 8 \times 10^5 (1 + 1.09))$ when the repayment is \$800 000 per annum. Find the value of B .

- ⓔ Henry borrowed \$100 000 at the beginning of 2007 at an annual interest rate of 6%. Each year the interest is calculated on the balance at the beginning of the year and added to the balance owing. What Henry owes is to be repaid in annual repayments of \$16 000, with the first repayment at the end of 2007.

- Ⓐ If A_n is the amount Henry owes after the n^{th} repayment, show that:

- Ⓘ the amount owing after the second repayment is $A_2 = 10^5(1.06)^2 - 16\,000(1 + 1.06)$.

Ⓙ $A_n = 10^5(1.06)^n - 16\,000\left(\frac{1.06^n - 1}{0.06}\right)$.

HINT

It is best to work out from first principles rather than using a formula. Also be careful in your simplifying of A_n .



- 8 In which year will Henry make his last payment?

- 9 An annuity is set up by paying \$30 000 every year with an interest rate of 9% per year.

- A The future value of this annuity at the end of n years can be found using $3 \times 10^4 \left(\frac{1.09^n - 1}{0.09} \right)$. Find the future value of this annuity paid at the end of 10 years. Give your answer correct to the nearest cent.

- B The present value of this annuity after n years can be found using $3 \times 10^4 \left(\frac{1.09^n - 1}{0.09(1.09)^n} \right)$. Find the present value of this annuity after 3 years. Give your answer correct to the nearest cent.

- 10 Find the smallest value of n , an integer, for which $2.5 \times 10^4 \left(\frac{1.06^n - 1}{0.06} \right) > 50\,000$.

- 11 Most items lose value with age. After n years the salvage value of an item bought for \$25 000 can be calculated using different formulae depending on what depreciation model is used. If the depreciation is arithmetic with, say, \$2 500 being lost every year, the salvage value will be $25\,000 - 2\,500n$. If the depreciation is geometric with, say, 10% being lost every year, the salvage value will be $25\,000(0.9)^n$.

- A After how many years will the items have equal salvage value using these two methods?

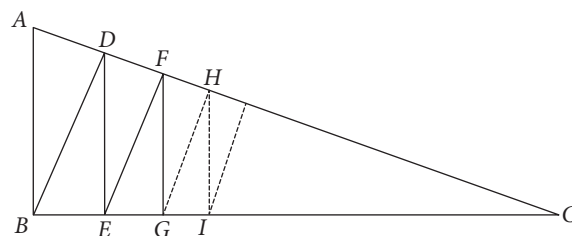
- B Which of the two methods has the item depreciating faster?

- 12 The triangle ABC has a right angle at B , side AB of length 4 units and $\angle CAB = \theta^\circ$. DE, FG, HI , etc. are all perpendicular to BC ; and BD, EF, GH , etc. are all perpendicular to AC . This process continues indefinitely.

- A Find the length of BD and DE .

- B Hence show that the length of FG is $4\sin^4\theta$.

- C Find the limiting sum (or sum to infinity) of $AB + DE + FG + \dots$



18 – Areas and integrals practice

All of the following functions have come from the idea of a probability density function. If your course includes probability density functions then these questions will give you practice with the mechanics of integration in those contexts. You may wish to rewrite the questions as probability questions.



THINGS TO KNOW

Because of the source of these questions, all the functions, $f(x)$, are continuous and the area between the graph of f and the x -axis, $\int_a^b f(x).dx$, is 1.

- ① A function $f(x)$ has a graph with the triangular form:

$$f(x) = \begin{cases} 1 + x & \text{for } -1 \leq x \leq 0 \\ 1 - x & \text{for } 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- Ⓐ Sketch the graph of f .

- Ⓑ Find the area under the graph between $x = -\frac{1}{2}$ and $x = +\frac{1}{2}$.

HINT

Perhaps you don't need to integrate.

- ② Find the exact value of the area under the graph of $y = f(x)$ for $x < \frac{1}{4}$:

$$f(x) = \begin{cases} 6x(1 - x) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{if } x < 0 \text{ or } x > 1 \end{cases}$$

SLIP

Careful of your manipulation of fractions.

- ③ For the function given by:

$$f(x) = \begin{cases} 12x^2(1 - x) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Ⓐ Find the exact value of the area under the curve for $x \leq \frac{2}{3}$.

TRICK

Multiply out first?

- Ⓑ Find the exact value of the area under the curve for $\frac{1}{3} \leq x \leq \frac{2}{3}$.

- ④ For the function of continuous variable, y

$$f(y) = \begin{cases} e^{-y} & \text{if } y \geq 0 \\ 0 & \text{if } y < 0 \end{cases}$$

find correct to 4 decimal places:

HINT

Integrating other functions needs practice too.



- A the area under the curve from $y = 0$ to $y = 1$.

- B the area under the curve between $y = 1$ and $y = 2$.

HINT

Draw the graph of the function to see where the negative cosine is likely to be.

- 5 The function of the continuous variable x is given by:

$$f(x) = \begin{cases} -0.5\cos x & \text{if } \frac{\pi}{2} < x < \pi \\ 0 & \text{elsewhere} \end{cases}$$

Find the value of k , correct to 1 decimal place such that the area under the curve for $x > k$ is 0.2.

- 6 The probability that I spend up to t hours on my homework is given by the area under the graph of the function $\int_0^t f(t)dt$ where $f(t) = \begin{cases} \frac{3}{2}t(t-1)^2 & \text{if } 0 \leq t \leq 2 \\ 0 & \text{for other values of } t \end{cases}$.

- A What is the probability that I spent

- I up to 1 hour on my homework on Monday?

- II more than 1 hour on my homework on Monday? _____

- B In the long run, I spent less than k minutes on my homework on $12\frac{1}{2}\%$ of nights. Find the value of k .

- C What is the probability, correct to three decimal places, that I spend longer than an hour on my homework on at least four out of five nights?

HINT

Careful – this one is more difficult.

19 – Tables of probabilities

Discrete probability distributions may not be in your syllabus so you should check with your teacher.
Of course, some of the questions can be done using common sense.



- 1 A movie star has a personal trainer who supervises her exercise every day. In the month of June the trainer’s records showed the number of push-ups done each day in the following table.

Number of push-ups each day (x)	0	10	20	30	40
Proportion of days on which x push-ups were done	$\frac{4}{30}$	$\frac{5}{30}$	$\frac{10}{30}$	$\frac{6}{30}$	$\frac{5}{30}$

- A On what proportion of the days of June did the movie star do at least 20 push-ups?
- _____
- B What was the average number of push-ups per day done by the movie star in the month of June?
- _____
- 2 Gen the librarian has been wondering whether the number of pages in a book affects how often the book is borrowed. From her records, she finds that 6.5% of borrowed books have about 125 pages.
- A Of the 4250 books borrowed this week, how many would she expect to have about 125 pages?
- _____
- B Of the 1050 books borrowed by the Year 9 class, how many would she expect to have **other than** about 125 pages?
- _____

Event	Probability of the event
Blonde	$3x^2 + 3x$
Brunette	$4x^2 + 2x$
Redhead	$2x^2 + 2x$
Mouse	x

- 3 In a particular town there are people with four different hair colours: blonde, brunette, redhead and ‘mouse’. The probabilities of these particular hair colours are shown in the table.
- A Knowing that the sum of the probabilities of these events is 1, show that x must be a solution to the quadratic equation $9x^2 + 8x - 1 = 0$.

SLIP

Simplify the sum carefully. Note that an exact value is asked for, so a fraction rather than a decimal answer is required. Do not round the expected (mean) value as it doesn’t have to be discrete.

- B Hence find the exact value of x .
- _____
- _____
- C Use substitution for the value of x to find the probability of a randomly selected person being a redhead.
- _____
- _____

The expected value of an event can be found by multiplying the probability of that event by the number of trials.

- D If a sample of 8 people is selected at random, how many would be expected to be redhead?
Give your answer correct to 2 decimal places.
- _____
- _____

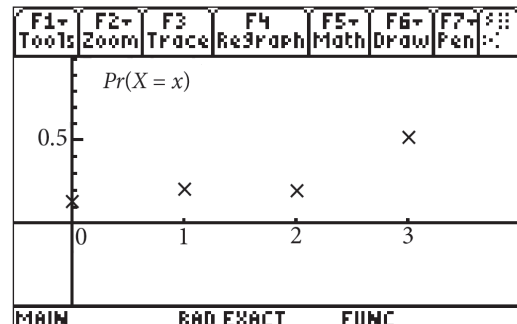


- 4 For the graph of probability $Pr(X = x)$ against the integer x , where $0 \leq x \leq 3$:

A state what the sum of the probabilities should be.

B complete the table of probabilities.

x	$Pr(X = x)$
0	
1	
2	
3	



C hence, plot the missing point.

D state the mode or the most probable value.

E find the mean or expected value.

F find the median or the half-way value.

- 5 For the probability distribution shown in the following table:

x	0	1	2	3	4
$Pr(X=x)$	0.2	0.3	0.15	0.25	0.1

A find the mean or expected value of x , $E(X)$;

B find the probability of an odd number.

C plot the graph of probability $Pr(X = x)$ against x , where $0 \leq x \leq 4$.

HINT

Don't join the points as the function is 'discrete'. Also make sure that your scale is clear and your plot is accurate.

20 – Mixed questions

It is sometimes useful to have practice in questions where the topics are mixed up.

This should ideally be done towards the end of the year when all the topics have been consolidated.



- ① When solving a pair of simultaneous equations for x and y , I substituted y from one equation into the other:

$$y - 2x = 7 \quad (1)$$

$$3x^2 - x - y + 1 = 0 \quad (2)$$

From (1): $y = 2x + 7$

in (2): $3x^2 - x - (2x + 7) + 1 = 0$

HINT

Use brackets!

- A Simplify this equation to a quadratic equation.

- B Factorise and solve it for x .

- C Substitute back into (1) to find solutions for y .

HINT

- ☞ Careful of brackets
- ☞ Take out the common factor
- ☞ Use null factor law
- ☞ Equation (1) is easier

- ② Is $(x - 2)$ a factor of $x^3 - x^2 - 14x + 24$? Give reasons for your answer. If so, find the other factors.

THINGS TO KNOW

This doesn't have numbers in it and you may find that more difficult. Think of the a and b as if they were numbers and look at the key words:

- ☞ 'linear' (from line graph) means that there are no squares, cubes etc. of x in the answer
- ☞ 'real field' and ' a and b are real positive numbers' means that you can have roots in the answer.

- ③ Find linear factors of $ax^4 - bx^2$ over the real field given that $a, b \in \mathbb{R}^+$.

- ④ For the function $f(x) = (x + 4)(x^2 - 4)$, find:

- A all the real solutions of the equation $f(x) = 0$.

- B the coordinates of the turning points of the graph of $y = f(x)$, correct to 2 decimal places.

- C the nature of the turning points of the graph of $y = f(x)$.

- D Hence on a separate sheet of paper, sketch the graph of $y = f(x)$, showing all turning points and intercepts on the axes.

HINT

Careful of your algebra at each step: expansions, factors, substitutions, equations and inequations. Remember that $f''(x) = 0$ is a necessary but not sufficient condition for a point of inflection. Make your graph large, neat and roughly to scale. Convex up is $\cap$.

SLIP

Careful of your algebra, particularly signs and brackets.



- E For what values of x is the graph of $y = f(x)$ convex up?

THINGS TO KNOW

Convex up is shaped $\cap$.

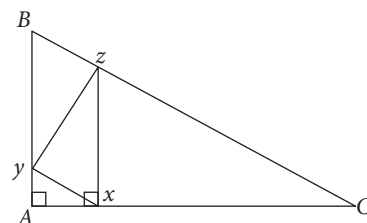
- 5 The rate of change of the number of viruses in a dish N at time t hours after they were placed there is proportional to the number of viruses in that dish at that time. N_0 viruses are placed in a dish at time $t = 0$ and the proportionality constant is k . Write down the exponential relationship between N and t .

THINGS TO KNOW

For an exponential function P with initial value of P_0 the rate of change of P is proportional to the value of P .

$$\frac{dP}{dt} = kP \text{ means } P = P_0 e^{kt} \text{ and } \frac{dP}{dt} = -kP \text{ means } P = P_0 e^{-kt}.$$

- 6 Triangle ABC is right-angled at A with side lengths AB of 5 cm and AC of 12 cm. Points X , Y and Z are shown in the diagram and are such that $\frac{AY}{AB} = \frac{BZ}{BC} = \frac{AX}{AC} = k$ and angle AXZ is a right-angle.



- A Find in terms of k the lengths of:

I AX. _____

II XC. _____

III ZX. _____

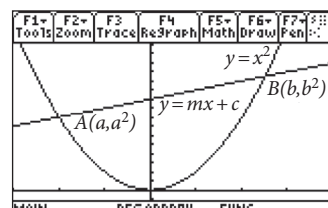
- B Hence find the area of triangle XYZ . _____

- C For what value of k will the area of triangle XYZ be greatest?

HINT

Think clearly and you won't lose your way. What is the height of triangle XYZ ? Will you use calculus or graphs or ...?

- 7 On the diagram shown, the parabola with equation $y = x^2$ is cut in two places $A(a, a^2)$ and $B(b, b^2)$ by a straight line with equation $y = mx + c$.



- A Copy and complete the following to show that $m = a + b$.

$A(a, a^2)$ is on the line with equation $y = mx + c$ so $a^2 = ma + c$

$B(b, b^2)$ is on the line with equation $y = mx + c$ so _____ (2)

Subtracting equations (1) – (2): $a^2 - b^2 =$ _____

Factorising: _____ = _____

$$m = \underline{\hspace{2cm}}$$

- B Copy and complete the following to show that $c = -ab$.

Substituting for m in equation (1): $a^2 =$ _____ $a + c$

Simplifying: _____ = _____

$$c = \underline{\hspace{2cm}}$$

HINT

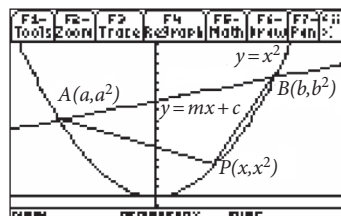
Remember that m and c are constants, not variables; only the x is a variable. Check the nature of the stationary point by some means.



- ③ For any point $P(x, x^2)$ on the parabola, it can be shown that the area of the triangle APB is given by

$$\frac{1}{2}(mx - x^2 + c)\sqrt{m^2 + 4c}.$$

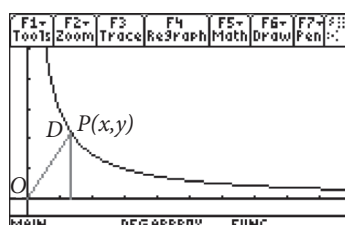
- ① Find the value of x in terms of m .



- ①① Hence find the coordinates of point P in terms of m so that the area of APB is a maximum.

- ⑧ A point P moves along the graph with equation $y = \frac{3}{x}$; $x > 0$. The distance of P from the origin, D , is shown in the diagram.

- ① Use Pythagoras' Theorem to show that $D^2 = x^2 + \frac{9}{x^2}$.



- ② Find the position of P for which the value of D^2 is a minimum.

- ③ Show that D^2 is a minimum when the line OP is perpendicular to this graph.

HINT

You need to explain where the $\frac{9}{x^2}$ comes from and make sure you answer the question! And make sure you confirm that it's a minimum. Hints: You want minimum D^2 , not D . You need to be careful of the algebra. Also, what is the condition for two lines to be perpendicular and what is the gradient of a curve?

- ⑨ A storage bin of volume 216π cubic metres is to be made in the shape of a cylinder of height h metres and radius r metres, which is topped by a hemisphere of radius r metres. The circular base is open.

- ① Draw and label a diagram showing the storage bin.

- ② The cost of constructing a square metre of the surface of the hemisphere is twice the cost of constructing a square metre of the surface of the cylinder. Show that the cost of constructing the bin can be given by $C = 4\pi r^2 + 2\pi rh$.

HINT

What is double what?

- ③ Use the volume of the bin to find a relationship between h and r .

- ④ Hence find the approximate values of r and h which will minimise the cost of constructing the storage bin.

Answers

Note – No answers are provided to questions such as 'show' and 'prove'.

1 – Key words and getting them right

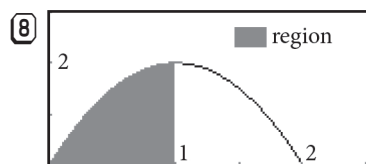
- ① (A) $\frac{9x-14}{12}$ (B) $2\frac{8}{9}$
 ② (A) $a - \frac{1}{a}$ (B) $a^2 - 2a - 1 = 0.1 \pm \sqrt{2}$ (C) 2.41, -0.41
 (D) $\frac{(a-1)(a+1)}{a} = \frac{1}{a}(a-1)(a+1)$
 ③ (A) $\frac{(x-5)}{x(x+5)}$ (B) 0 (C) $\frac{-9 \pm \sqrt{41}}{4}$
 ④ square both sides: $4(x+7) = 9x^2$; expand LHS: $4x + 28 = 9x^2$;
 group on one side: $9x^2 - 4x - 28 = 0$; factorise:
 $(9x + 14)(x - 2) = 0$; $x \geq -7$ so under square root isn't negative
 ⑤ (A) DOCS $(x-3y)(x^2 + 3xy + 9y^2)$ (B) DOPS $(a-3b)(a+3b)$
 (C) CF & DOPS $x^2(x-3)(x+3)$ (D) Quadratic $(2x-1)(x+2)$
 (E) CF & Quadratic $x(x-4)(x-2)$ (F) FTh $(3x+1)(x-1)(x-1)$
 (G) CF & FTh $x(x-1)(x-3)(x+2)$
 (H) Group in pairs $(x^2+2)(x-4)$ (I) CF $2b(a-b)$
 (J) Quadratic $(3x-5)(x+2)$ (K) FTh $(x+1)(x-2)(x-3)$
 (L) DOPS & simplify $3(1+x)(1-x)$
 Spelling list: no answer given

2 – Tricks and traps in integration

- ② $\log_e(x^2 + 1)$ + any constant or no constant
 ③ (A) $x^3 - \frac{3}{4}x^4 + 9x^2 - 5x + C$ (B) $\frac{1}{15}e^{15x} + C$ (C) $-\frac{1}{15}\cos 15x + C$
 (D) $\frac{1}{3}e^{3x} - 3\log_e(x+1) + \frac{2}{3}x^{\frac{3}{2}} + C$
 ④ $2x - \tan x$
 ⑤ (A) $15\log_e x + C$ (B) $\frac{-1}{x} + c$
 ⑥ (A) $\frac{2\sqrt{3}}{3}x^{\frac{3}{2}} + C$ (B) $2\log_e(x-3) + C$ (C) $-x^3 - 4x^2 + C$
 ⑦ $\frac{9}{2}x^6 - 9x^5 + 6x^4 - \frac{4}{3}x^3 + C$

3 – Definite Integrals

- ① (A) $\frac{64}{3}$ (B) $\frac{1}{5}(e^5 - 1)$
 ② $\frac{\sqrt{3}}{4}$
 ③ $\frac{1}{3}\log_e 5$
 ④ $\sqrt{3}$
 ⑤ 400.656
 ⑦ (A) $2\frac{2}{3}$ (B) $2\frac{2}{3}$ units²; graph is above the x-axis for all values of x
 so there are no issues of signed area to consider.



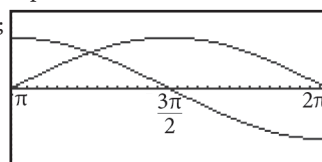
- ⑨ Area of a semicircle is $\frac{\pi r^2}{2} = 2\pi$
 ⑩ 78.134 or 78 thousand litres
 ⑪ (A) -6, 3 (B) 0

4 – Integration and area under the curve

- ① (B) unit²
 ② expansion: $x^3 + 2x^2 - 5x - 6$; $21\frac{1}{12}$
 ③ less than since the integral gives the signed area
 (above the axis – below the axis)
 ④ notation can use absolute value or sign: $-\int_p^q f(x)dx + \int_q^r f(x)dx$
 ⑤ (A) 36 (B) $\left[-\frac{1}{3}x^3 - 3x^2\right]_0^k$ and result will be -36
 since $\int_{-6}^0 (-x^2 - 6x)dx + \int_0^k (-x^2 - 6x)dx = 0$.
 ⑥ (A) $(x+2)(x^2 - 2x + 1) = (x+2)(x-1)(x-1)$ (B) repeated solution
 means the graph touches the x-axis
 (C) $\frac{27}{4} = 6.75$
 ⑦ $\frac{9\pi}{4} - \int_1^3 (-x^2 + 4x - 3)dx$
 $\log_e 3$
 ⑧ (A) $\log_e 3$ (B) $\log_e 3 - \int_0^{\log_e 3} e^x dx$
 ⑨ (A) $\int_0^{\sqrt{3}} [(8-2x^2) - (x^2-1)]dx = \int_0^{\sqrt{3}} (9-3x^2)dx$
 (B) $\int_0^{\sqrt{3}} [(8-2x^2) - (x^2-1)]dx = \int_1^{\sqrt{3}} (x^2-1)dx$
 ⑩ π square units ≈ 3.142 square units

5 – Rates of change in calculus

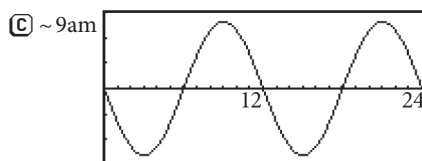
- ① 0
 ② $-\frac{2}{\pi}$
 ③ 2.87
 ④ 17.18; 27.18
 ⑤ (A) $\frac{1}{4}$ (B) $\frac{1}{b}$
 ⑥ $2 + \frac{15}{x^2}$
 ⑦ -4.853 m³ per sec
 ⑧ $-\cos(k)$;



- ⑨ (A) metres per second
 ⑩ (A) m²/s (B) 125.66 m²/s (C) 130 m²/s

6 – Rates of change in practice

- ① 18 L per 100 km to 2 sig figs
 ② -1.6 cm³/s; negative because volume is reducing.
 ③ (A) $10 + \log_e(1.25) \approx 10.223$ hours (B) 6.181 gm per hour
 ④ (A) $5000/9 \approx 555.56$ litres (B) 90 minutes
 (C) -37.04 litres per minute
 (D) at the beginning when the drum is full.
 ⑤ (A) $0 \leq t \leq 0.628(\frac{\pi}{5})$; P is a periodic function so values repeat
 (B) $10a\cos 10t$ (iii) 60 units
 ⑥ (A) $h'(t) = -\frac{5\pi}{6}\sin \frac{\pi t}{6}$ (B) -2.267 m/h; falling



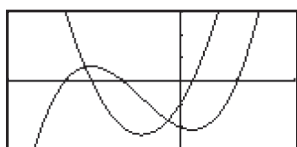
- 7 (A) mms^{-1} (B) 157.1 mms^{-1}
 8 (A) $-e^{-3t}(3\cos 2t + 2\sin 2t)$ (B) -0.028 m/s

7 – More than a rate

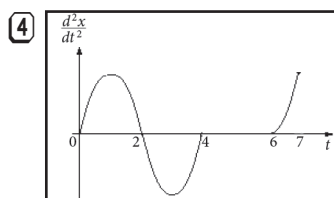
- 1 (A) cm^3 per second (B) cm per second (C) cm^3 per cm
 2 (A) (i) $\frac{dr}{dt} = p$ (ii) $\frac{dV}{dt} = 4\pi r^2 p$ (B) $36\pi p$
 3 (A) $300x^{\frac{1}{2}}$ (B) $\frac{dx}{dt} = \frac{dx}{dy} \cdot \frac{dy}{dt}$ (C) $\frac{dx}{dt} = \frac{1}{30x}$ (D) $\frac{1}{90}$
 4 (A) $\frac{dx}{dt}$ (B) $\frac{dv}{dt}$; positive (increasing) (C) $\frac{dx}{dt} = \frac{dx}{dv} \cdot \frac{dv}{dt}$
 (D) $\frac{dx}{dv}$ (E) $v = 108x + x^3$ (F) $108 + 3x^2$ (G) 0.1 (H) cm/s
 5 $\frac{dr}{dt} = -\frac{1}{12\pi} \text{ cm per min}$ (or -0.027 cm/min)
 6 (A) Use $r = \frac{1}{2}h$ in volume formula (B) 1.53 m/min

8 – Moving particles

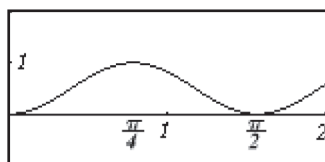
- 1 (A) metres
 2 C: displacement vs time, A: velocity vs time, B: acceleration vs time
 3 $-5.4\pi \sin(\frac{3\pi}{2}t + \frac{\pi}{4}) = 0.5$



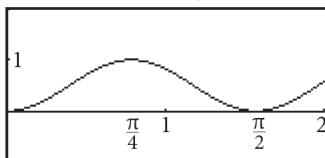
Stationary points (a maximum and a minimum) correspond to zeros on the gradient function.



- 5 Area under the graph is the distance travelled as long as the velocity doesn't go negative.
 6 (A) $t = 7$ (B) $t = 2$; $t = 4$; area under curve is zero



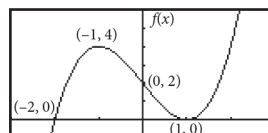
- 7 (A) 5 sec (B) 5 sec; distance travelled is maximum
 8 (A) Use $v = 0$ at $t = 0$ to get $C = \frac{1}{2}$
 (B)



(C) $\frac{\pi}{2} \approx 1.57$ (D) $\int_0^{\frac{\pi}{2}} (-\frac{1}{2}\cos 4t + \frac{1}{2}) dt = \frac{\pi}{4} \approx 0.785$
 9 (A) $t^2 + \frac{8}{\pi^2} \cos \frac{\pi t}{2} + c$ (B) $-\frac{8}{\pi^2}$ (C) $t^2 + \frac{8}{\pi^2} \cos \frac{\pi t}{2} - \frac{8}{\pi^2}$
 (D) $9 - \frac{8}{\pi^2}$ (E) $9 - \frac{8}{\pi^2}$ metres

9 – Integrals and graphs

- 1 the curve because the area under $f(x)$ increases quickly then slowly...
 2 (A) $f(x) = x^3 - 3x + 2$ (B) Max: $(-1, 4)$; Min: $(1, 0)$; x -intercepts $(-2, 0)$, $(1, 0)$, y -intercept $(0, 2)$



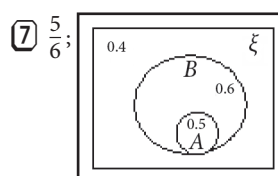
- 3 For reference $h'(x) = f(x) = 12x^3 - 12x^2 - 12x + 12$

$h(x)$ increasing, decreasing, stationary	$h'(x) = f(x)$	Range of values of x
increasing	positive	$-1 < x < 1$ and $x > 1$
decreasing	negative	$x < -1$
stationary	zero	$x = -1$ and $x = 1$

- 4 $(0, k)$
 5 A3, x -axis; A2, y -axis; B1, x -axis; B4, y -axis;
 6 (A) 0.5 (B) $1.3, 1, 1.3, 2.7$ (C) 3.15 unit^2
 7 (A) $1.875, 2, 2.125, 3$ (B) 4.56 unit^2 (C) 4

10 – Probability and events

- 1 (A) independent (B) independent (C) mutually exclusive
 (D) mutually exclusive (E) dependent (F) not mutually exclusive
 2 (A) 0.2 (B) 0.45
 3 0.7
 4 (A) $\frac{3}{4}$ (B) $\frac{5}{8}$ (C) $\frac{7}{8}$
 5 (A) 0.2 (B) 0 (C) 0.2
 6 $\frac{3}{4}$



- 8 (A) $\frac{1}{12}$ (B) $\frac{1}{6}$ (C) $\frac{5}{6}$ (D) $\frac{1}{6}$ (E) $\frac{1}{2}$ (F) $\frac{1}{2}$
 9 (A) $\frac{100}{49} \approx 0.0671$ (B) $\frac{398}{447} \approx 0.8904$
 10 0.45
 11 (A) $0.6^3 = 0.216$ (B) 0.144 (C) 0.268



11 – Tables and counting

① $\frac{2369}{3764}$

② (A)

	Red card	Blue card	Total
Number divisible by 4	0	2	2
Number not divisible by 4	5	3	8

(B) (i) $\frac{1}{2}$ (ii) $\frac{1}{4}$ (iii) $\frac{7}{10}$ (iv) 0

③ (A)

	Number the test indicated had drunk some alcohol	Number the test indicated had not drunk some alcohol	Total
People who had drunk some alcohol	80	20	100
People who had not drunk alcohol	260	40	300

(B) 120 (C) 30% (D) $\frac{7}{10}$ (E) 0.2

④ 0.225

⑤ (A) 7! (B) $\frac{4!3!}{7!} = \frac{1}{35}$ (C) $\frac{4!4!}{7!} = \frac{4}{35}$

⑥ (A) $6! = 720$ (B) $4!3! = 144$ (C) $\frac{1}{5}$ (D) $\frac{1}{5}$ (E) $\frac{1}{10}$

12 – Discrete probability questions

① (A) $\Pr(X=2) = \binom{5}{2} (0.12)^2 (0.88)^3 = 0.098$

(B) $\Pr(X \geq 2) = 1 - [\Pr(X=0) + \Pr(X=1)] = 0.112$

② (A) $\Pr(X=2) = \binom{8}{2} \left(\frac{20}{81}\right)^2 \left(\frac{61}{81}\right)^6 = 0.311$

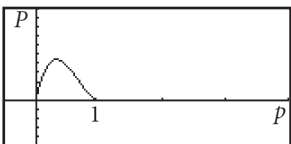
(B) $\Pr(X=0) = \binom{n}{0} \left(\frac{20}{81}\right)^0 \left(\frac{61}{81}\right)^n \leq 0.09, n=9 (8.49)$

③ (A) (a) 0.4 (b) 4 (B) 6 black and 9 white

④ (A) 0.812 (B) 0.07

⑤ 0.4613

⑥ 24 (23.43)

⑦ (A) $P = 3p(1-p)^2$ (B)  (C) $\frac{1}{3}$

⑧ (A) $\frac{1}{5}$ (B) (i) 0.000786 (ii) 0.000864

13 – Probability and selections

① (A) 0.0526 (B) 0.9474

② (A) $\binom{99}{2} = {}^{99}C_2 = 4851$ (B) 0.04

③ $\frac{47}{95}$

④ $\frac{41}{55} \approx 0.7455$

⑤ (A) 0.0365 (B) 0.224 (C) 0.963

⑥ $k=1$ only since $k=6$ and 9 don't apply to both cases.

⑦ 3.77×10^{-8}

⑧ (A) $\frac{196}{5513} \approx 0.0356$ (B) $\frac{196}{5513} \approx 0.0356$ (C) $\frac{4950}{11026} \approx 0.4489$

⑨ (A) 3 red and 7 black (B) no (C) $\frac{7}{15}$ (D) $\frac{7}{15}$ (E) $\frac{2}{3}$ (F) $\frac{2}{9}$

⑩ the five of spades; a five; a spade; a heart; not the five of spades.

14 – Normal distribution

① 0; 0; 0

② $-\frac{7}{4}; \frac{7}{4}$

③ 0.9249

④ (A) 0.4772 (B) 0.9772 (C) 0.0228 (D) 0.0228 (E) 0.8413 (F) 0.1574; 3

⑤ (A) $0 \leq x \leq 24$ (B) 0.9974

⑥ 0.2525

⑦ 61 (60.36)

⑧ 0.7942

⑨ 43.8 cm

⑩ (A) (i) 0.4309 (ii) 0 (B) 0.12(085)

15 – Coordinate geometry

① negative

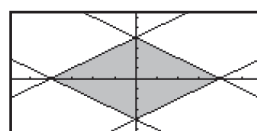
② (A) $5\sqrt{2} \approx 7.0711$ (B) $(-\frac{1}{2}, \frac{1}{2})$ (C) -1 (D) 1 (E) $y = x + 2$

③ C; E

④ $y = -\frac{1}{3}x + \frac{7}{3}$

⑤ -48.43° or 131.57°

⑥ Area is a square side length $\frac{1}{4}\sqrt{2}$ so area is $\frac{1}{8}$ units²



⑦ (A) $2\sqrt{2}$ units; $2\sqrt{2}$ units (B) product of gradients -1 or angle in a semicircle (C) 4 square units

⑧ (A) $A(-3, 5); B(2, 0)$ (B) $y = -x + 2$

⑨ (A) $(5, 2\frac{1}{2})$ (B) $\frac{\sqrt{29}}{2}$ units (C) $AB \parallel DC$ and $BC \parallel AD$ because gradients are the same

⑩ (8, 0)

16 – Geometry and trigonometry

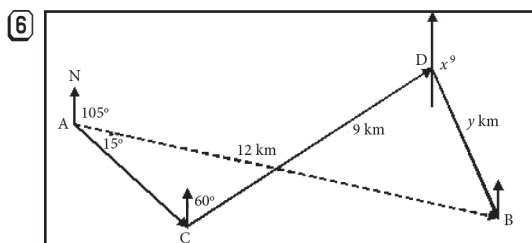
① (A) ABCD (B) APD (C) APD (D) use parallelogram, $\angle ADC = 30^\circ$, cointerior angles, equal angles of isosceles triangle, trapezium or other

② (A) 100.29° (B) 20.66 cm^2

③ (A) 134.6° (B) 20.8°

④ (A) SAS ($AY = YC$, right angles at Y, XY common) (B) all 60° (C) 30° (D) 1:2 or $\frac{1}{2}$

⑤ (A) 54 cm (B) 53.19 cm (C) 2430 cm^2



- 7 (A) 0.6972 $\approx$ 0.70 (B) 0.5 (0.4769)
 8 (A) 2.1 (B) 1.5 (C) 100° (D) 3.2 (E) 26.8°

17 – Sequences and series

- 1 (A) 75 (B) 150 (C) (i) 524 288 (ii) 9 (D) $n = 9$; 4.122 (E) 1.5
 2 7.867
 3 $r = \sin\theta$; $\frac{4\sin\theta}{1-\sin\theta}$
 4 72.5 km
 5 (A) 72 000; 72930.4 (B) 3rd term
 6 (A) E: \$50 000; L: \$49 920; Eleanor (B) \$66 000 (C) \$68 318.97
 (D) E: \$570 000; L: \$576 293.14; \$6293.14
 7 \$4 268 500
 8 (A) (ii) use $A_n = 10^5 (1.06)^n - 16\,000(1 + 1.06 + 1.06^2 + \dots + 1.06^{n-1})$
 (B) 8.06 years; 2015 (still some left at the end of 2014)
 9 (A) \$455 787.89 (B) \$75 938.84
 10 2
 11 (A) $n = 1$ (B) arithmetic
 12 (A) $4\sin\theta$; $4\sin^2\theta$ (B) pattern of sequence
 (C) $a = 4$; $r = \sin^2\theta$; $= \frac{4}{1-\sin^2\theta} 4\sec^2\theta$ (simplification depends on the syllabus)

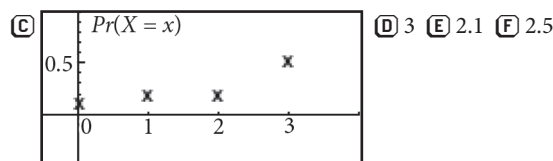
18 – Areas and integrals practice

- 1 (A) (B) $\frac{3}{4}$
 2 $\frac{5}{32}$
 3 (A) $\frac{16}{27}$ (B) $\frac{13}{27}$
 4 (A) 0.6321 (B) 0.2325
 5 2.5
 6 (A) (i) $\frac{1}{8}$ (i) $\frac{7}{8}$ (B) 60 (C) 0.879

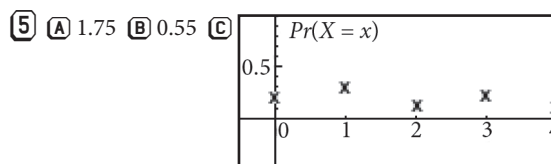
19 – Tables of probabilities

- 1 (A) $\frac{7}{10}$ (B) 21
 2 (A) 276.25 (B) 981.75
 3 (A) Sum of probabilities $9x^2 + 8x = 1$ (B) $\frac{1}{9}$ (C) $\frac{20}{81}$ (D) 1.98
 4 (A) 1 (B)

x	$Pr(X=x)$
0	0.1
1	0.2
2	0.2
3	0.5

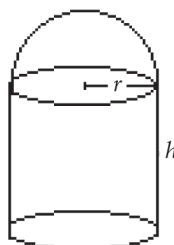


(D) 3 (E) 2.1 (F) 2.5



20 – Mixed questions

- 1 (A) $3x^2 - 3x - 6 = 0$ (B) $3(x-2)(x+1) = 0$ so $x = 2, -1$
 (C) $y = 11, 5$
 2 Yes, factor theorem $P(2) = 0$; other factors $(x+4)(x-3)$
 3 $x^2(\sqrt{ax} - \sqrt{b})(\sqrt{ax} + \sqrt{b})$
 4 (A) $x = -4, 2, -2$ (B) and (C) (0.43, -16.90) minimum;
 (-3.10, 5.05) maximum
 (D) (E) $x < 0.43$
 5 $N = N_0 e^{kt}$
 6 (A) (i) 12k (ii) $12 - 12k$ (iii) $5(1-k)$ (B) $30k(1-k)$ (C) $k = 0.5$
 7 (C) (i) $x = \frac{1}{2}m$ (ii) $(\frac{1}{2}m, \frac{1}{4}m^2)$
 8 (A) $D^2 = x^2 + y^2$ (B) $(\sqrt{3}, \sqrt{3})$
 (C) gradient of OP is 1, gradient of the curve is -1
 9 (A)



- (B) Use surface areas of hemisphere and open cylinder and include cost

(C) $h = \frac{216}{r^2} - \frac{2r}{3}$ (D) $r = \sqrt[3]{81} \approx 4.327$; $h \approx 8.654$

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